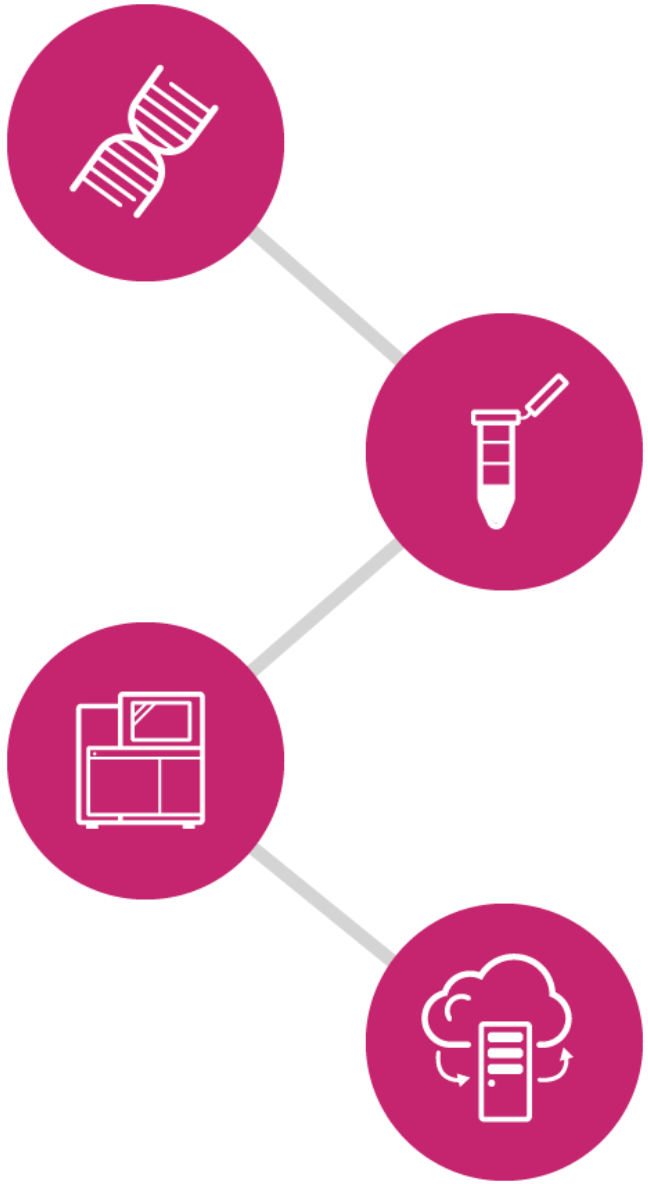
Sample to Data

Yavuz Ariyurek

Leiden Genome Technology Center

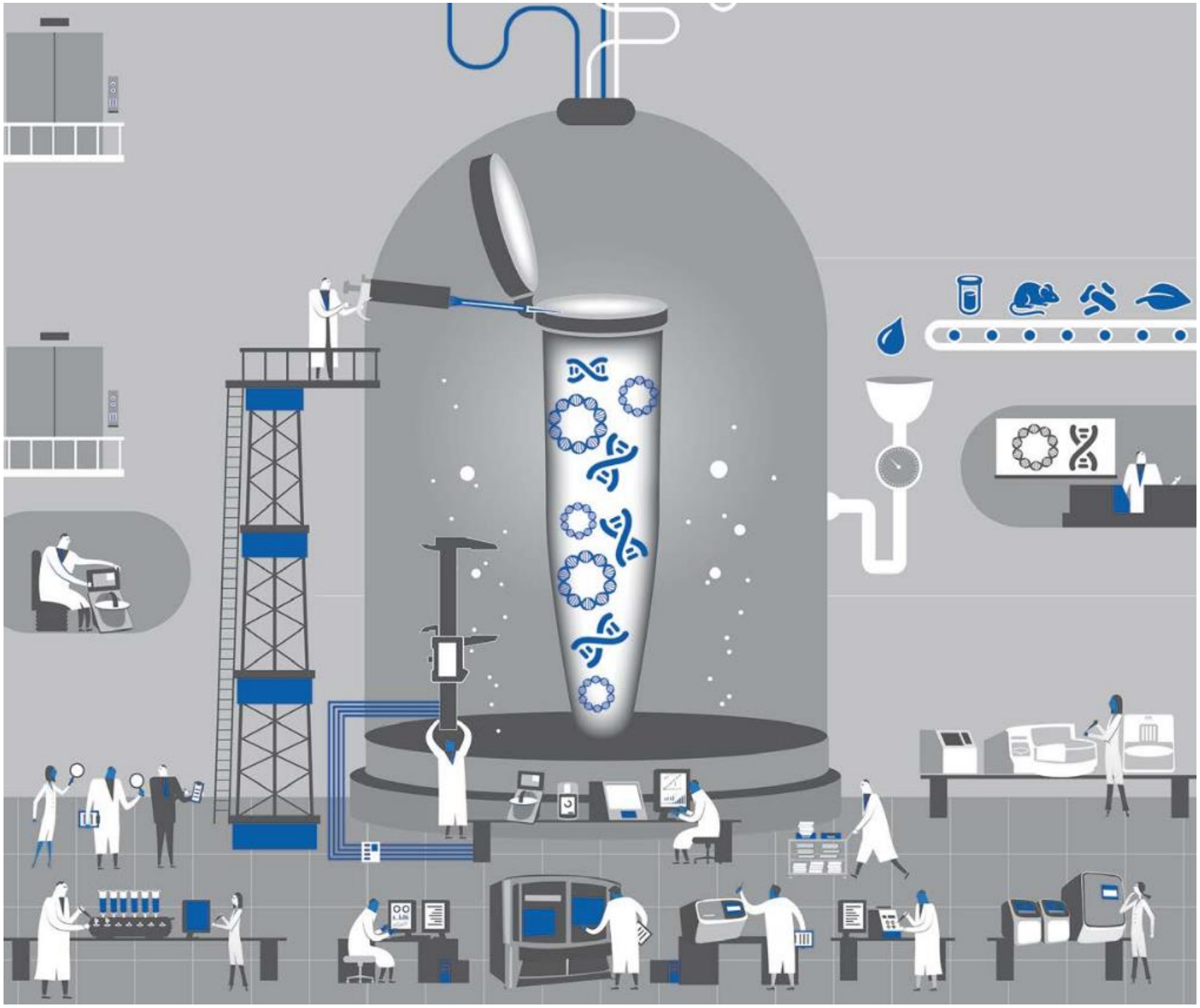
LUMC

03 November 2025



The (real) scientific method.





What can be sequenced?

Whole genome DNA

- Shotgun libraries
- Whole genome amplification (WGA)

Targeted DNA

- Capture / whole exome sequencing (WES)
- Amplicon sequencing / gene panel

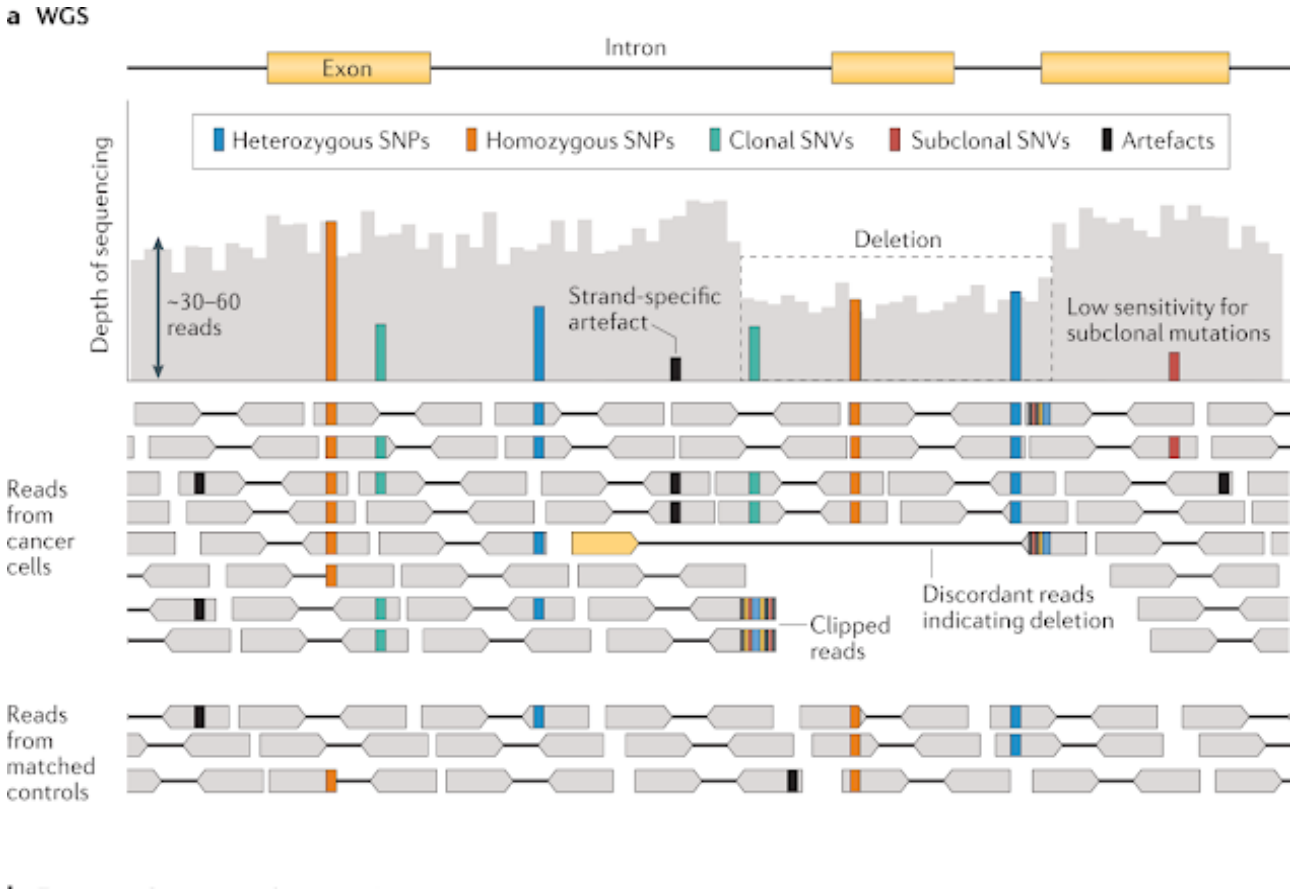
Bulk RNA

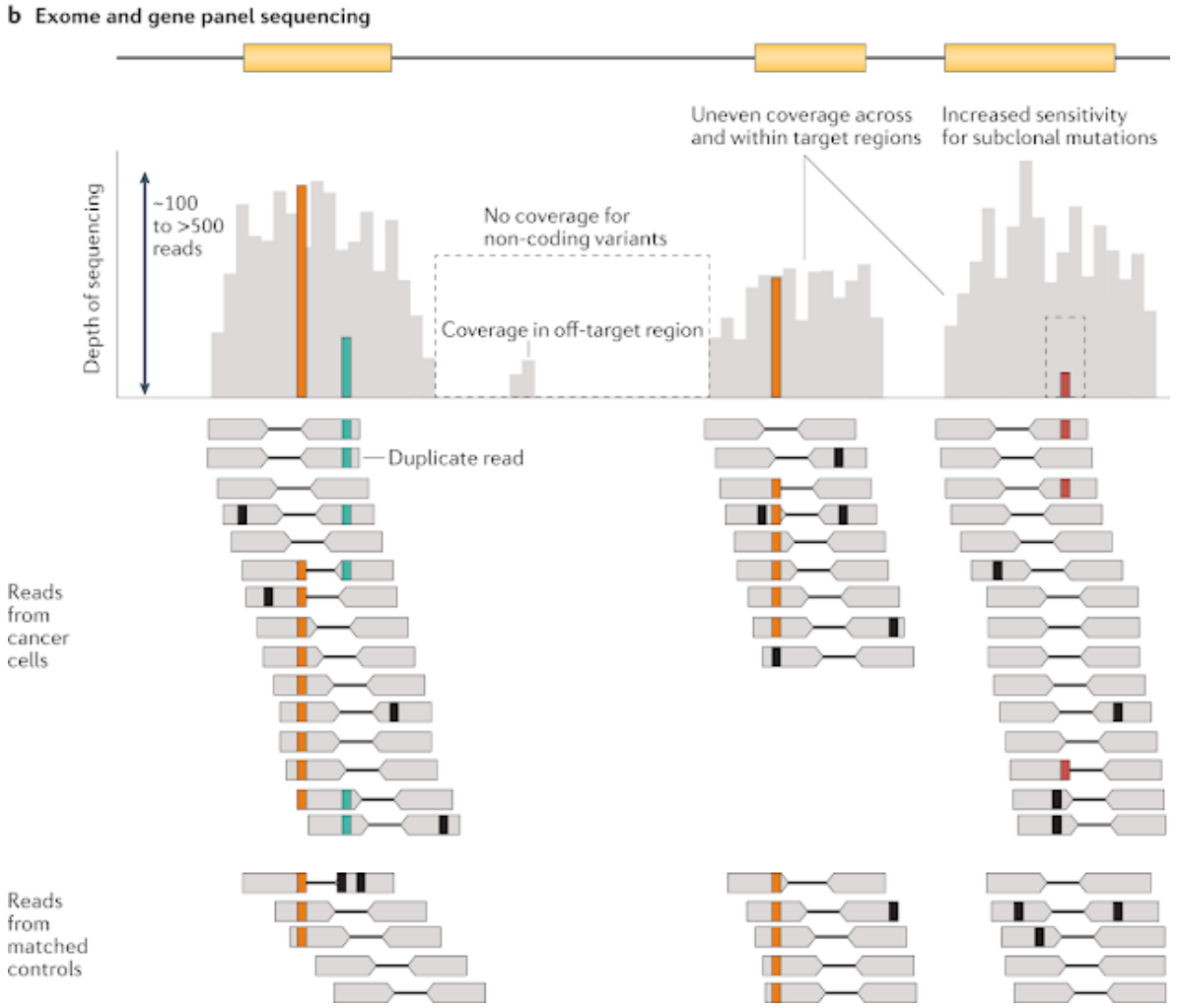
- Direct
- cDNA synthesis

Single-cells

- RNA
- DNA (ATAC-seq, Methyl-seq, copy number variation)

Variant Detection – short reads

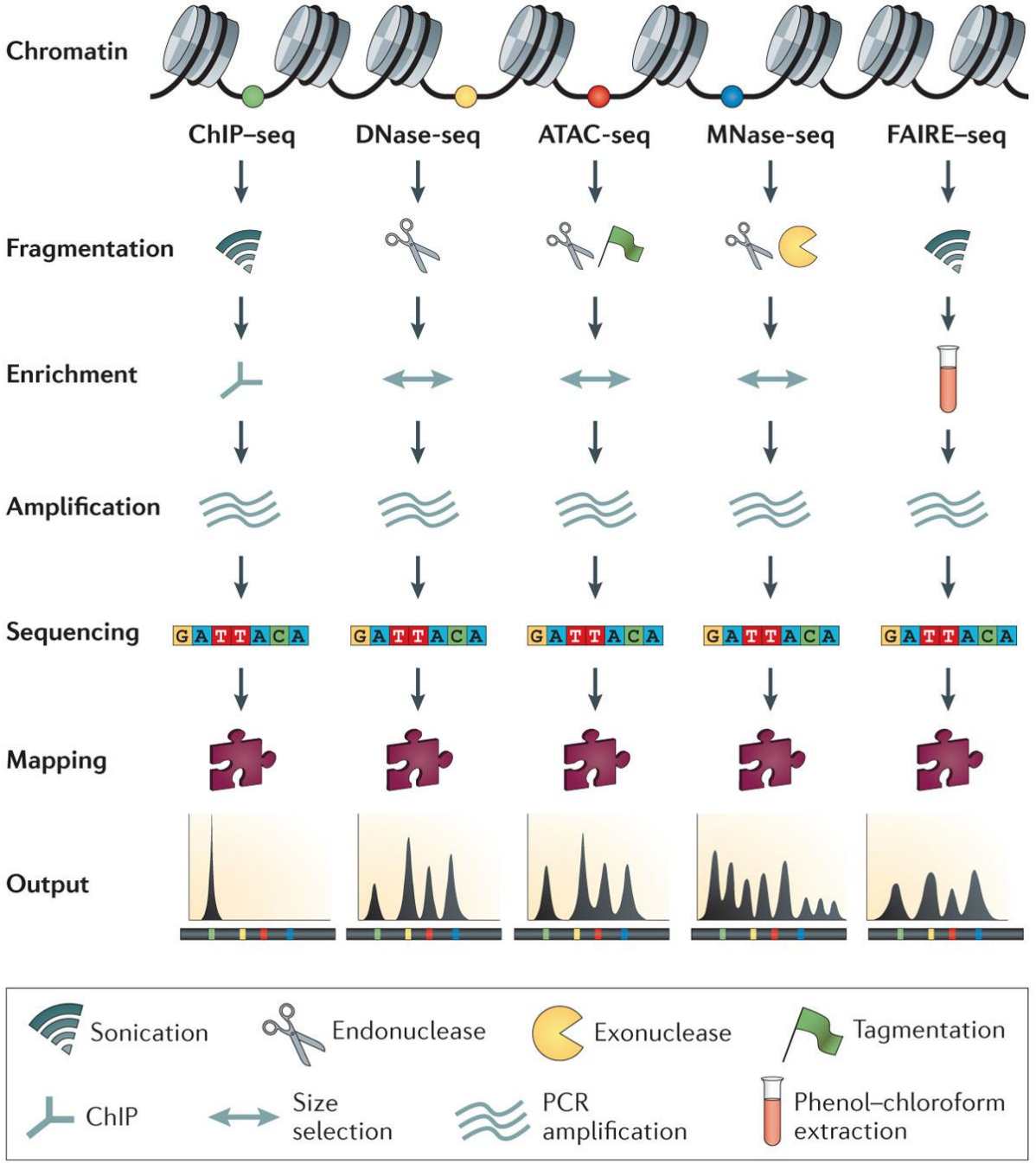




Germline or Somatic?
DNA/RNA Modifications included
Denovo Assembly

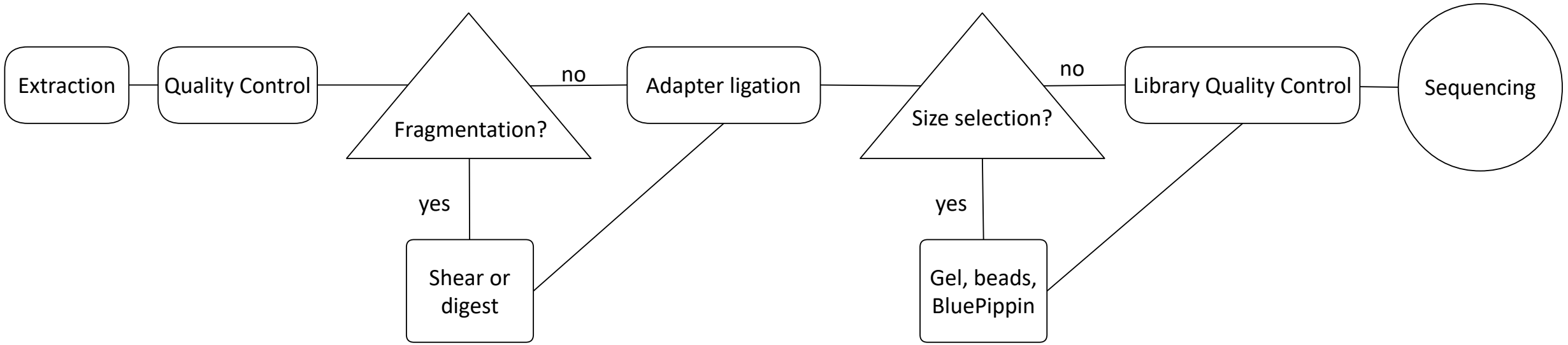
Peak Detection - example

RNA-seq
smallRNA
ATAC-seq
CopyNumberVariation
..



https://www.illumina.com/content/dam/illumina-marketing/documents/applications/ngs-library-prep/for-all-you-seq-single-cell.pdf

Sample prep workflow



DNA extraction

- Extraction = breaking down the extracellular matrix and opening the cell membranes using enzymes, solvents, or surfactants. The DNA in the resulting mixture must then be isolated.

- Phenol – gold standard but tricky to work with
- Spin columns – easy, but purified DNA is short
- Gravity-based column – DNA is >50kb but slower than spin column
- Chloroform – alternative to phenol
- Agarose plug – requires single-cell suspension
- Magnetic beads – easy to automate

DNA QC

Check for inhibitors of enzymatic reactions

Sample-specific: Hemoglobin, bile salts

Extraction-specific: EDTA, heparin, phenol:chloroform

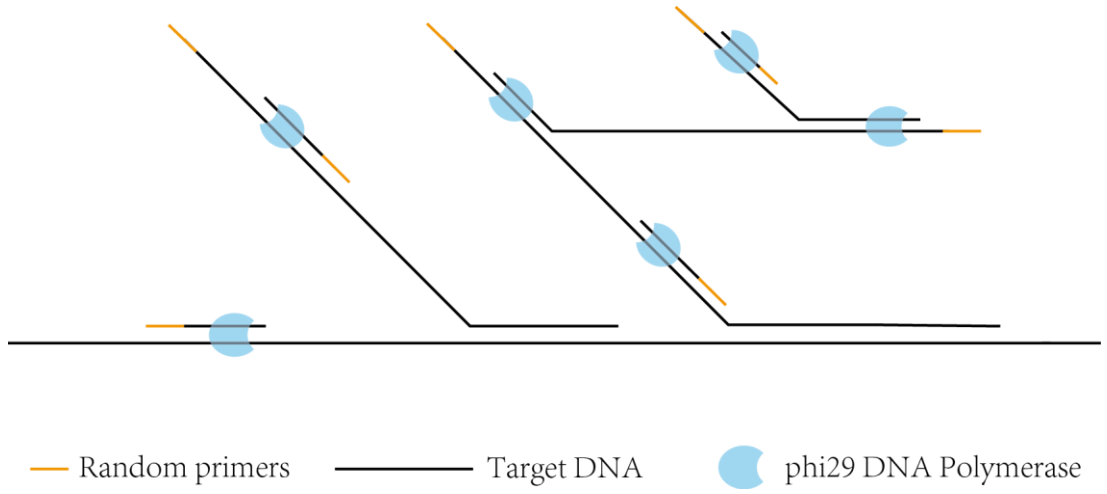
Use UV spectrophotometry to assess the purity of the DNA

Substance	Absorbance	260/280 Value	260/230 Value
Pure DNA	260 nm	~ 1.8	2.0-2.2
EDTA, Carbs, Phenol	230 nm	< 1.5	< 2.0
Guanidine HCl	230 nm	< 1.5	< 2.0

Use fluorometric-based methods for quantitation (Qubit, Pico/RiboGreen)

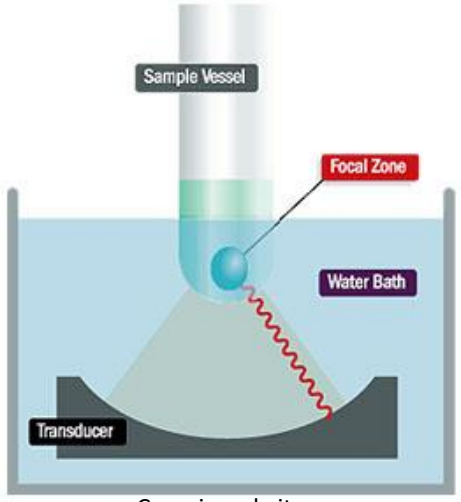
DNA amplification

- Shotgun libraries: >1ug starting material, high quality, skip this step
- Inputs <1ug may need amplification
  - PCR – targeted amplification of 1-100s of loci
  - MDA – multiple displacement amplification, “unbiased” whole genome amplification (WGA) utilizing random hexamers, better for SNVs
  - MALBEC – better for copy number variation



DNA fragmentation 200bp-1000bp

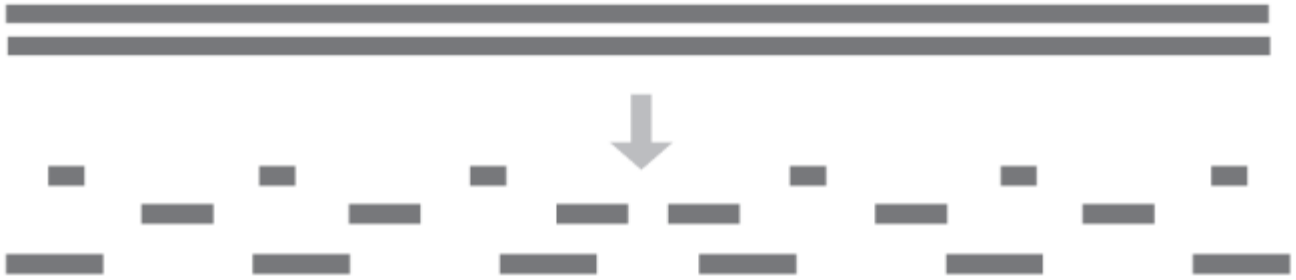
- Shotgun libraries must be fragmented before Illumina or Ion Proton sequencing, but not for PacBio or ONT sequencing
- Mechanical or enzymatic?
  - Mechanical shearing – uses sound or columns to randomly fragment DNA
  - Enzymatic fragmentation – no upfront investment, just a tube of enzyme, but suffers from preferential cleavage



Covaris website

Mechanical shearing

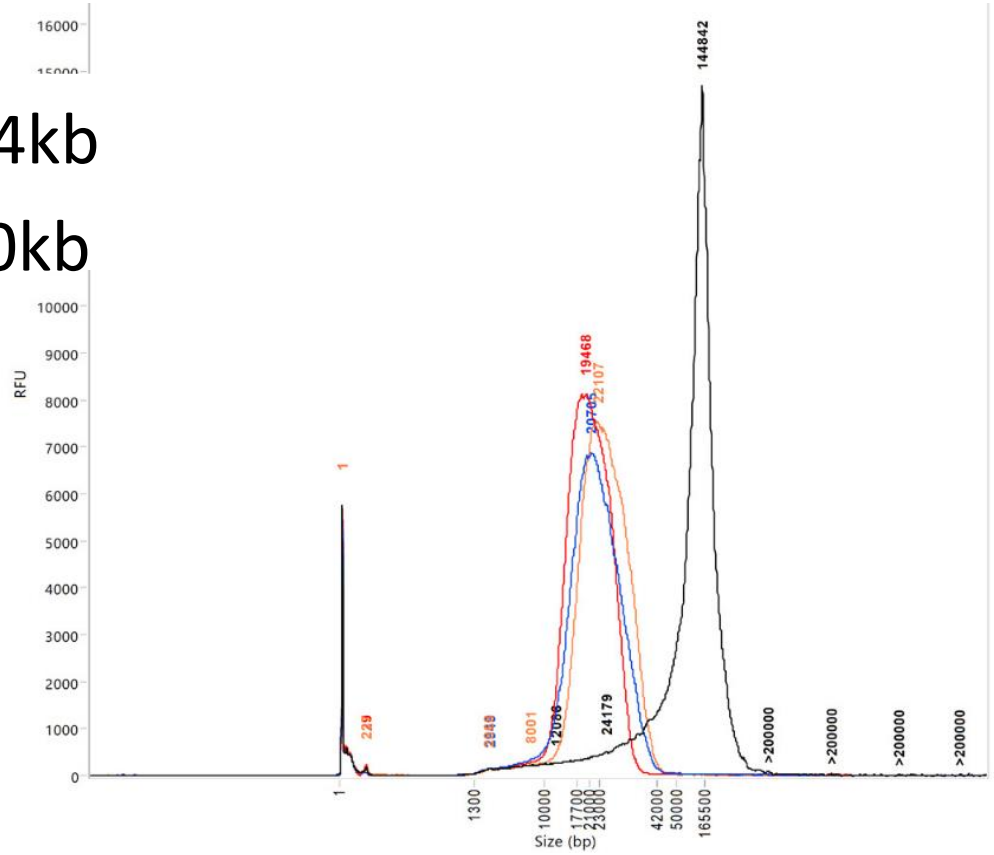
Enzymatic fragmentation



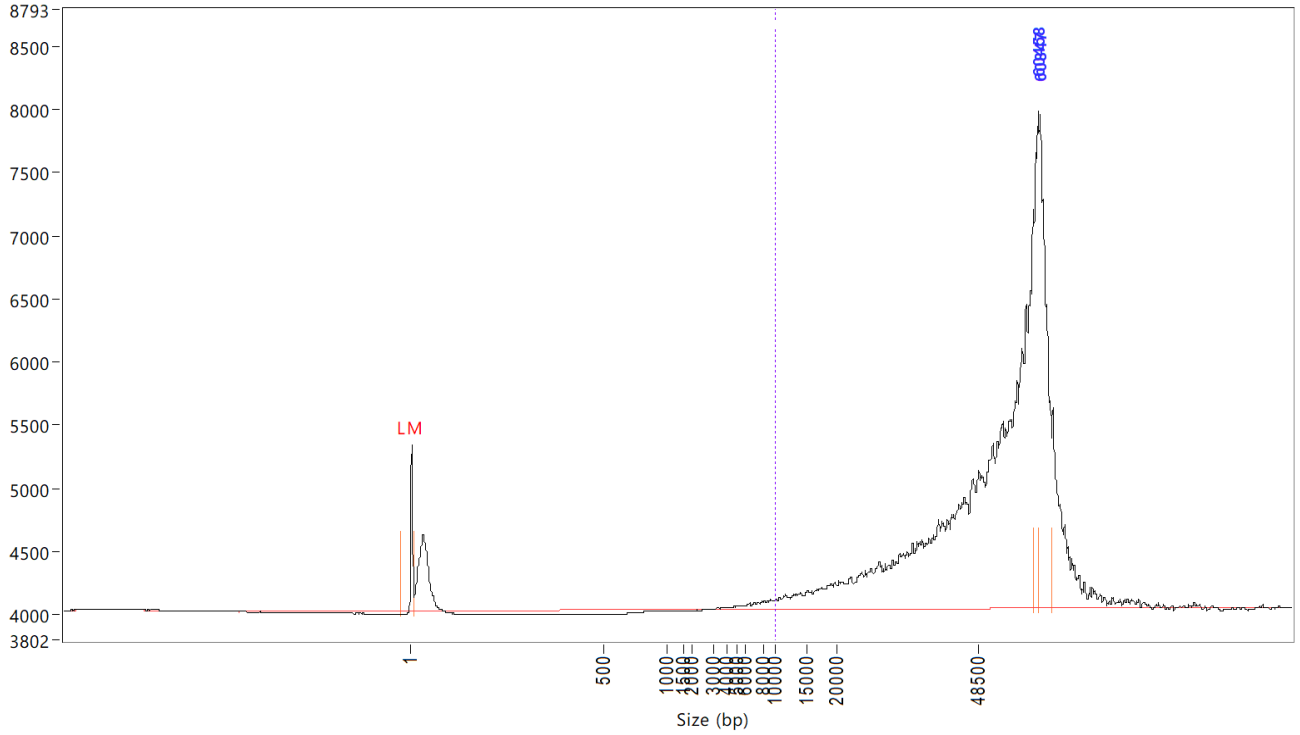
Illumina website

DNA fragmentation Long Read

- Depends on platform or quality of gDNA
- Is Ultra Long Reads or phasing needed?
- PacBio ideal insert size for ccs reads: 20-24kb
- ONT ideal insert size for production: 20-40kb
- Mechanical shearing:
  - Megaruptor
  - Needle shearing
  - High Throughput DNA shearing using Automation



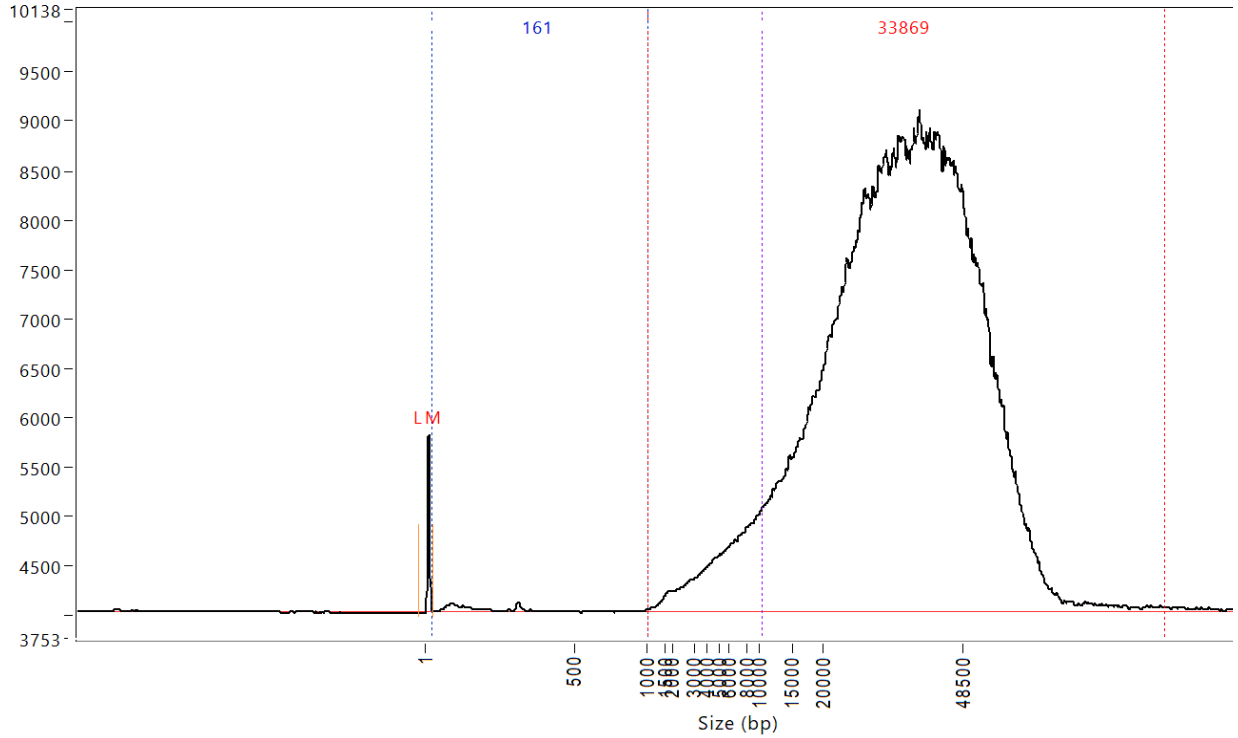
gDNA quality varies (1)



Good quality gDNA

PacBio: needs shearing

ONT: needs shearing unless Long reads are required

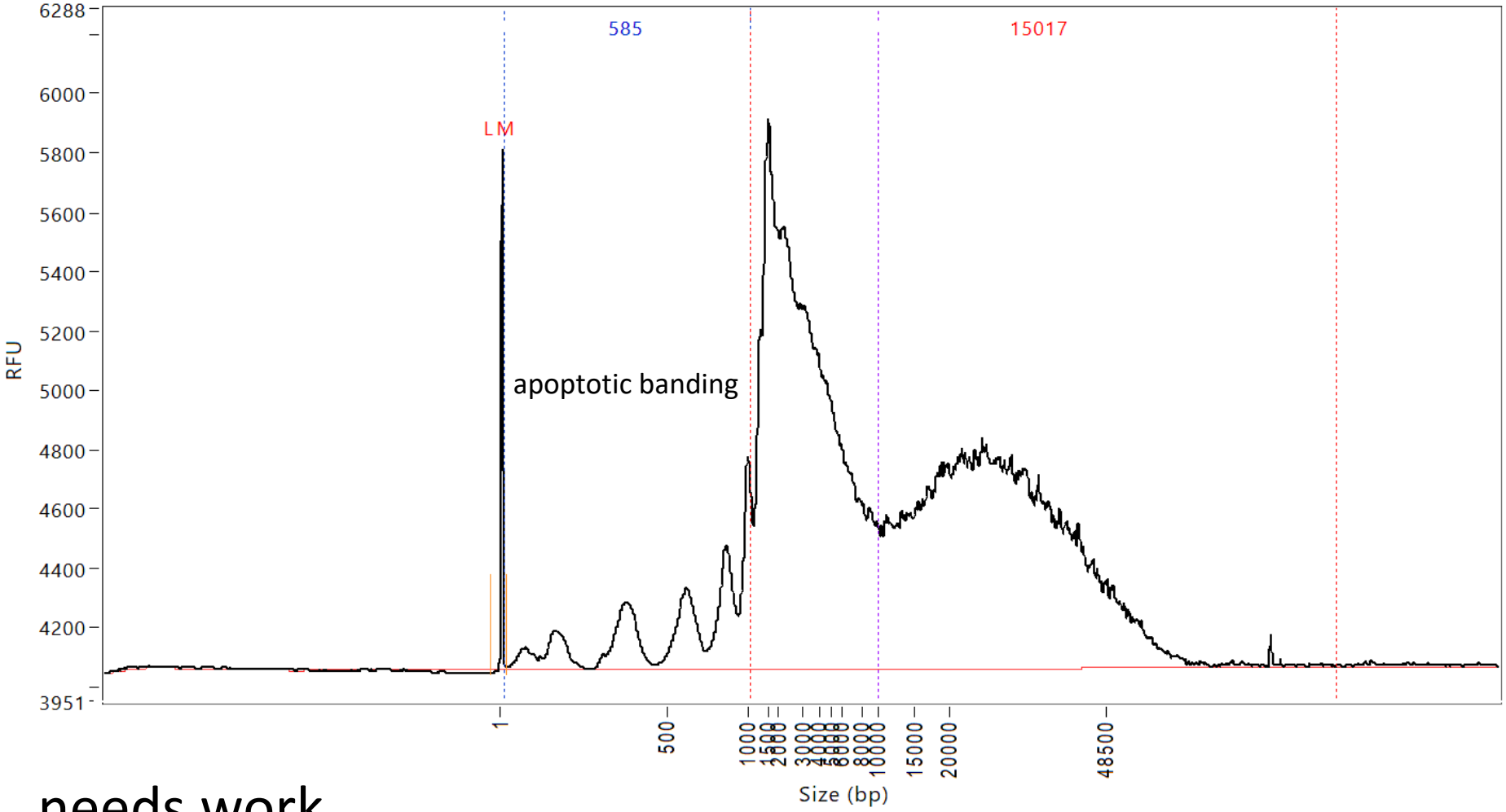


Good quality gDNA

PacBio: needs shearing

ONT: needs Size Selection to increase N50

gDNA quality varies (2)



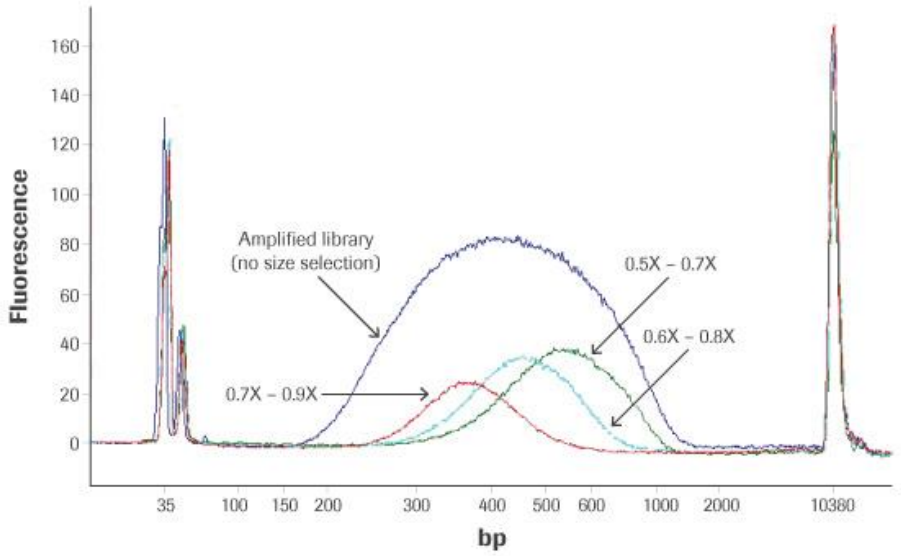
needs work

PacBio: needs SizeSelection and shearing

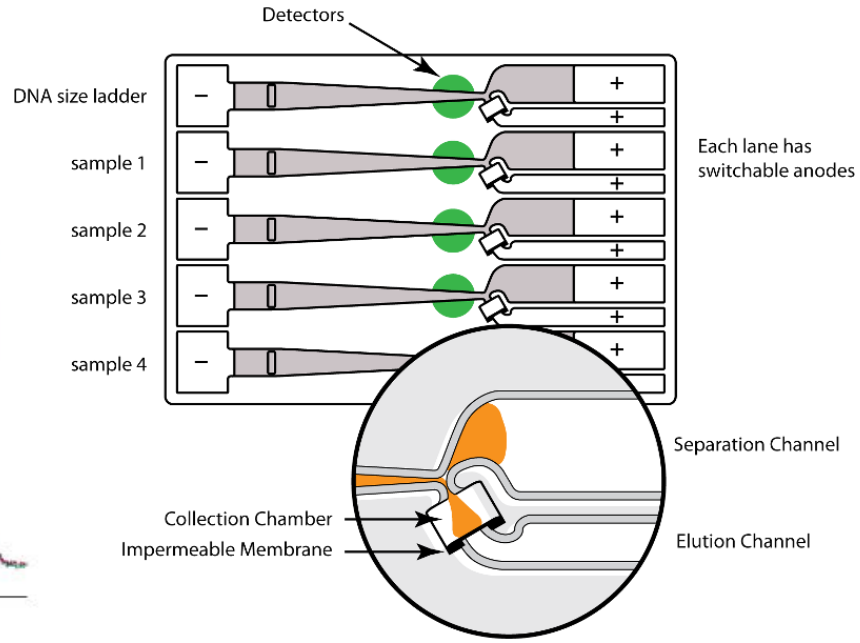
ONT: needs Size Selection

DNA size-selection

- For short-read sequencers, a narrow size range is required (200-700bp)
- For long-read sequencers, we want to remove small fragments
- Gel purification
  - Agarose
  - BluePippin
- Bead purification
  - XP beads



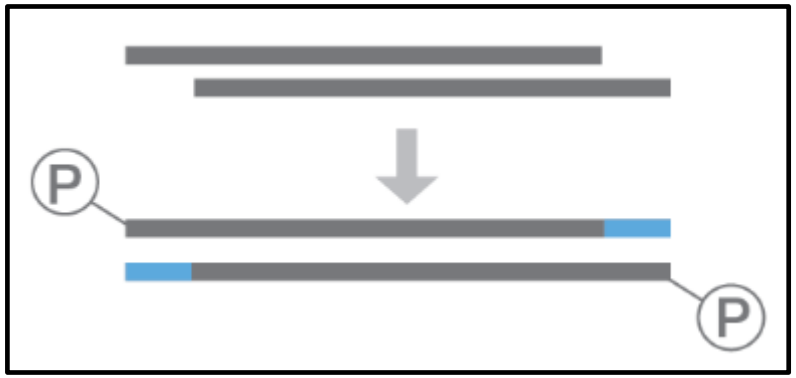
Roche website



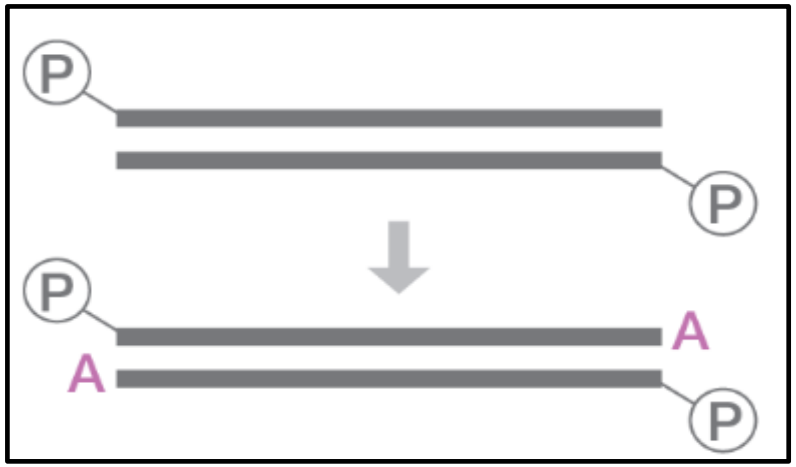
Sage science website

Adapter ligation - Illumina

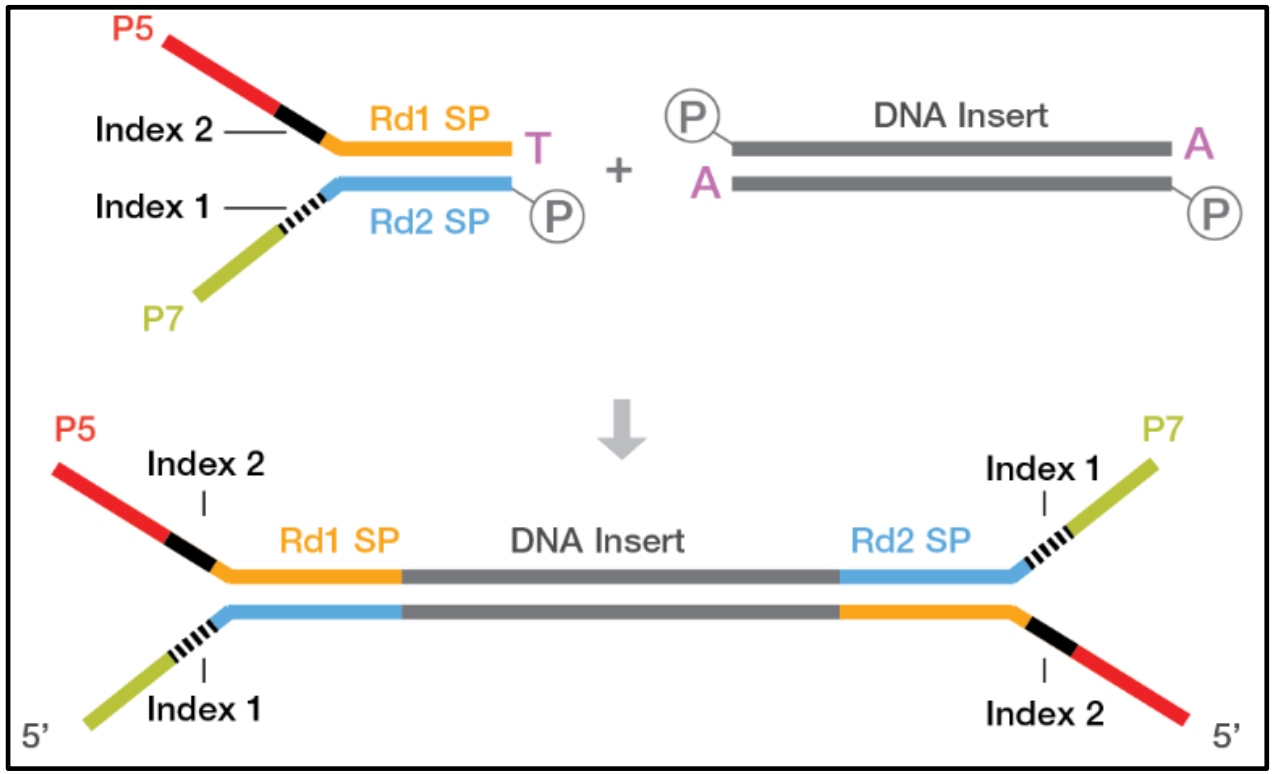
1. End Repair



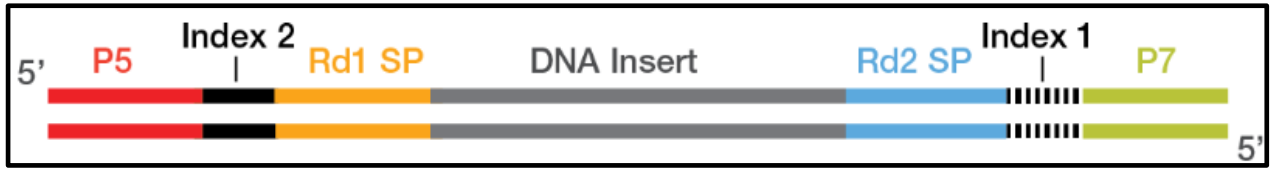
2. A-tailing



3. Ligate index adapter

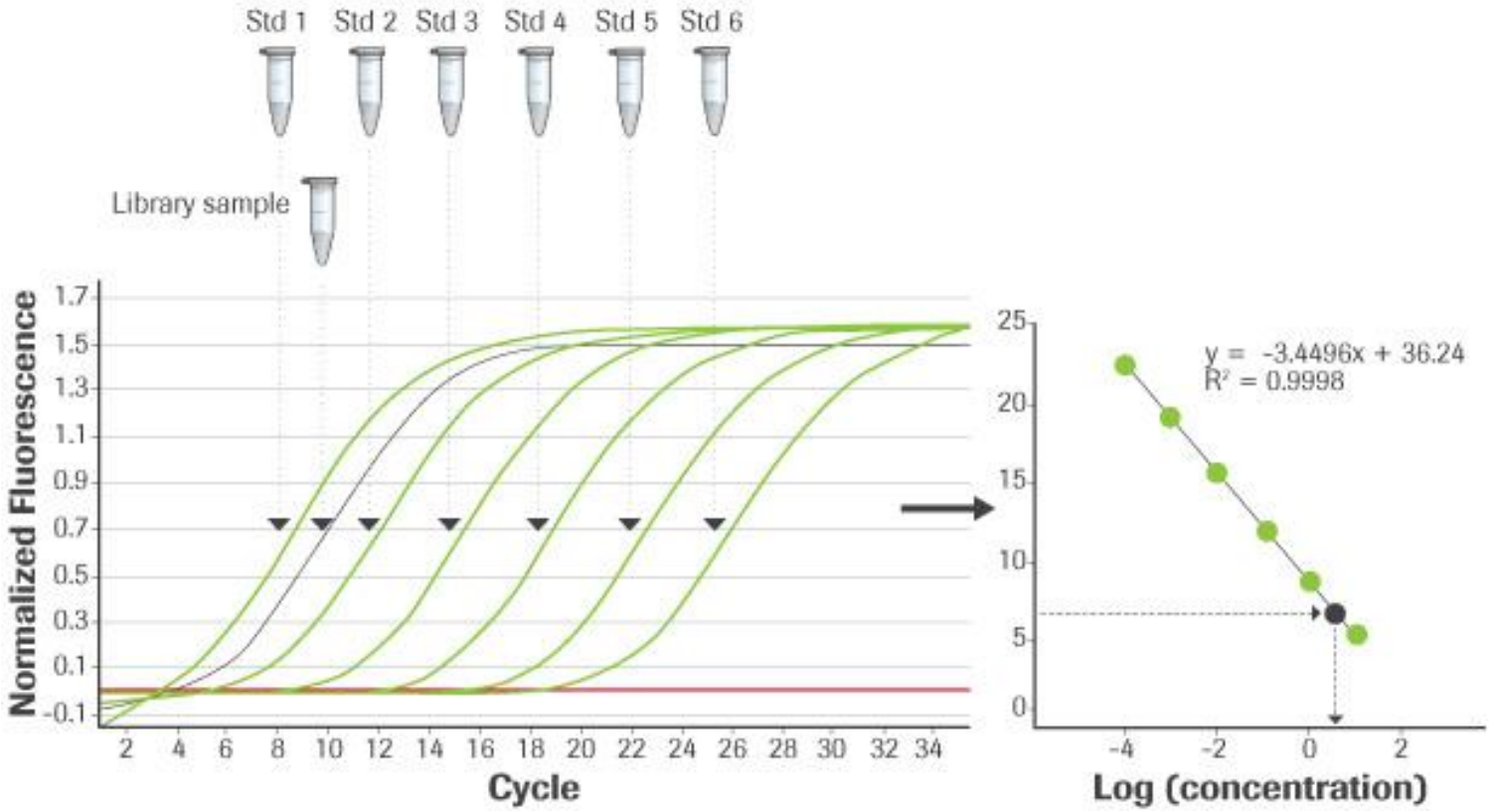


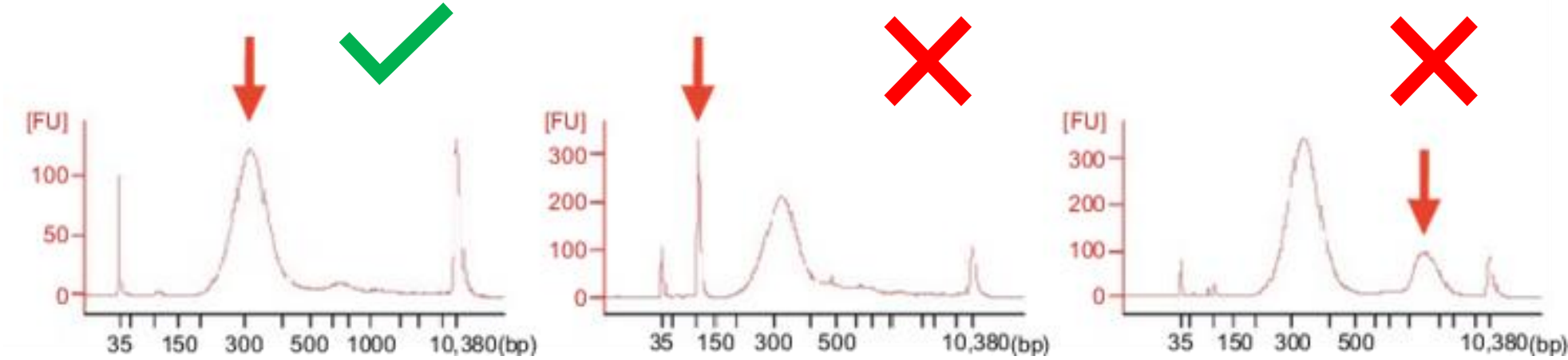
4. Amplification



DNA Library QC

- Quantity
  - Fluorescence
  - qPCR
  - ddPCR
- Quality
  - Bioanalyzer
  - Femto Pulse

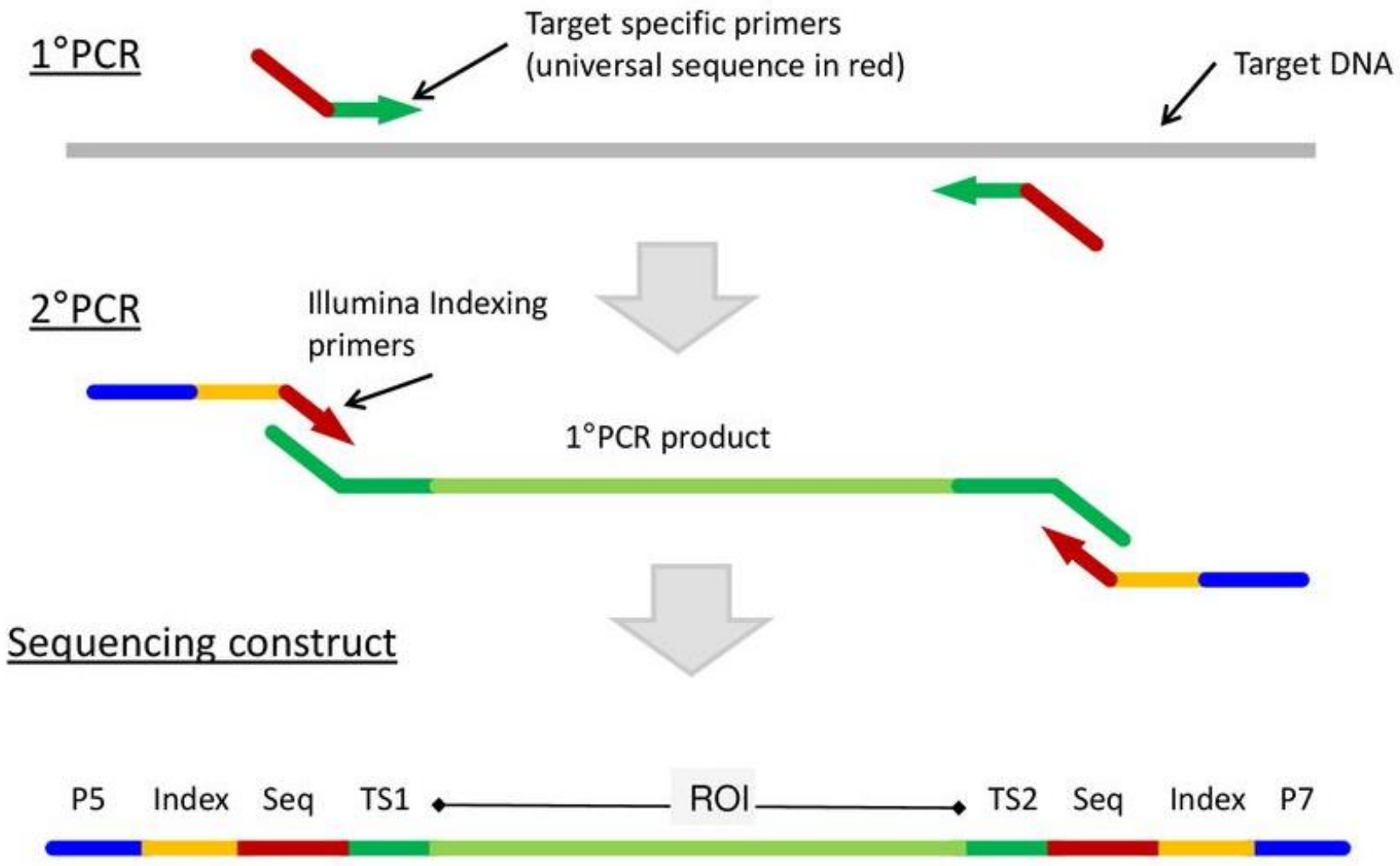




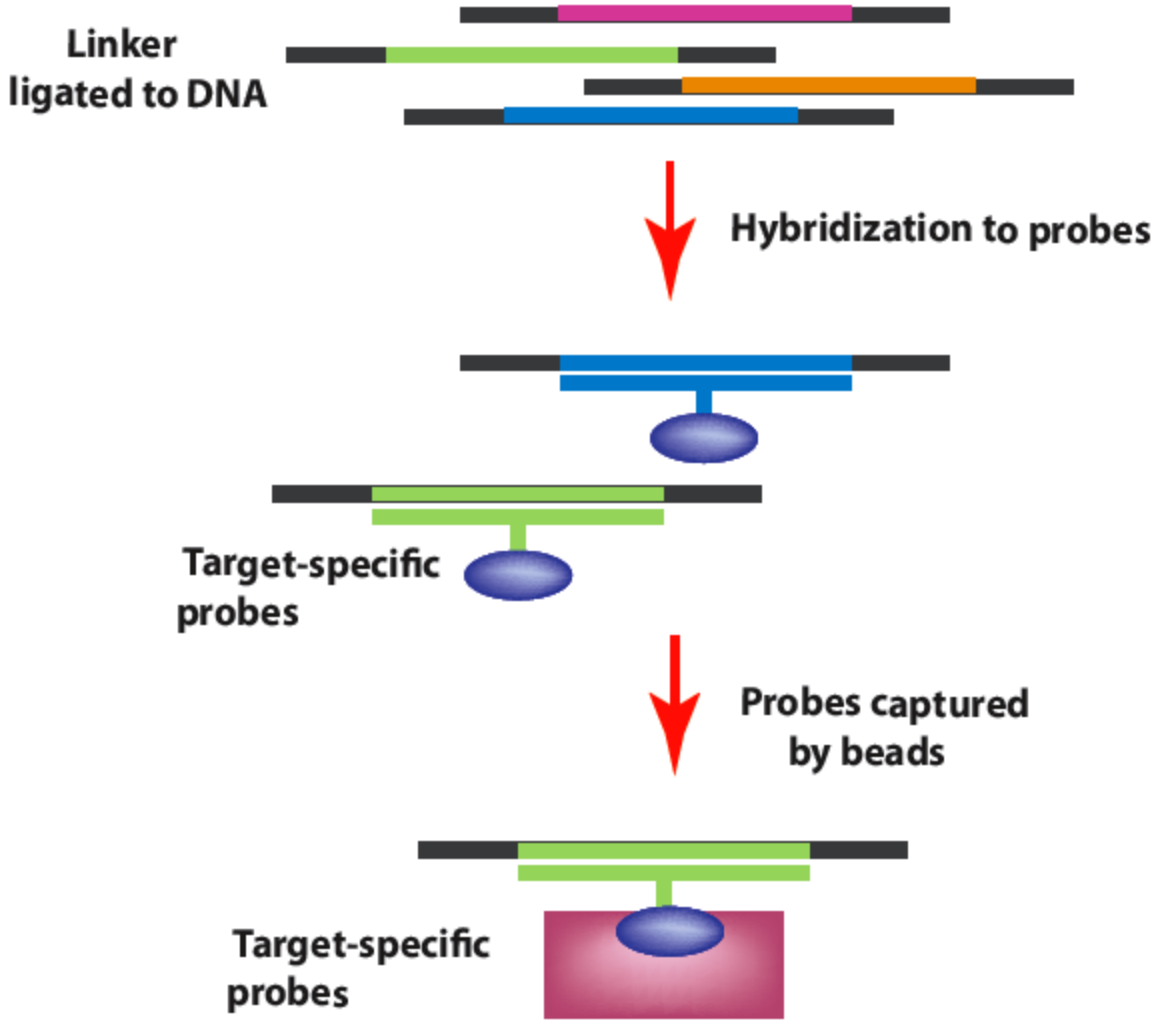
Targeted DNA library prep

- Capture – Whole exome sequencing (WES)
- Amplicons – PCR-based amplification of 1-100s of loci
- CRISPR-Cas9 – Amplification-free targeted excision of small regions of the genome
- Adaptive Sampling – Enrichment during sequencing (ONT only)

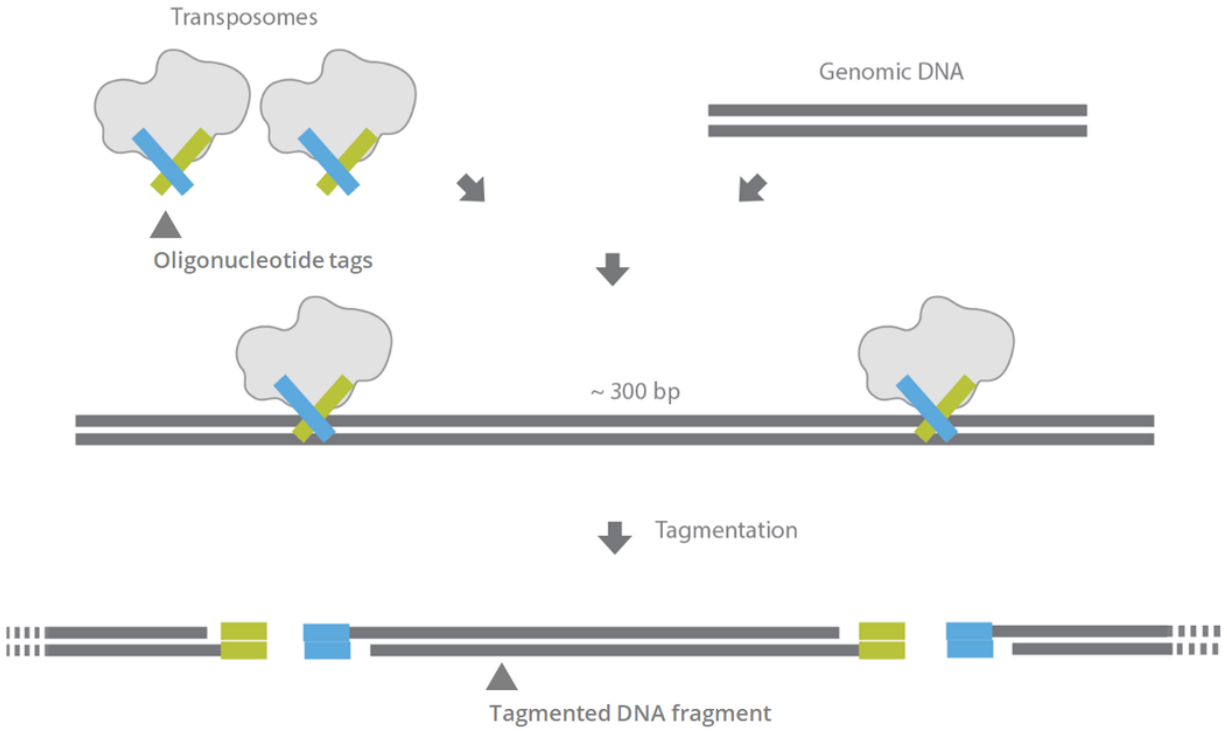
Amplicon sequencing

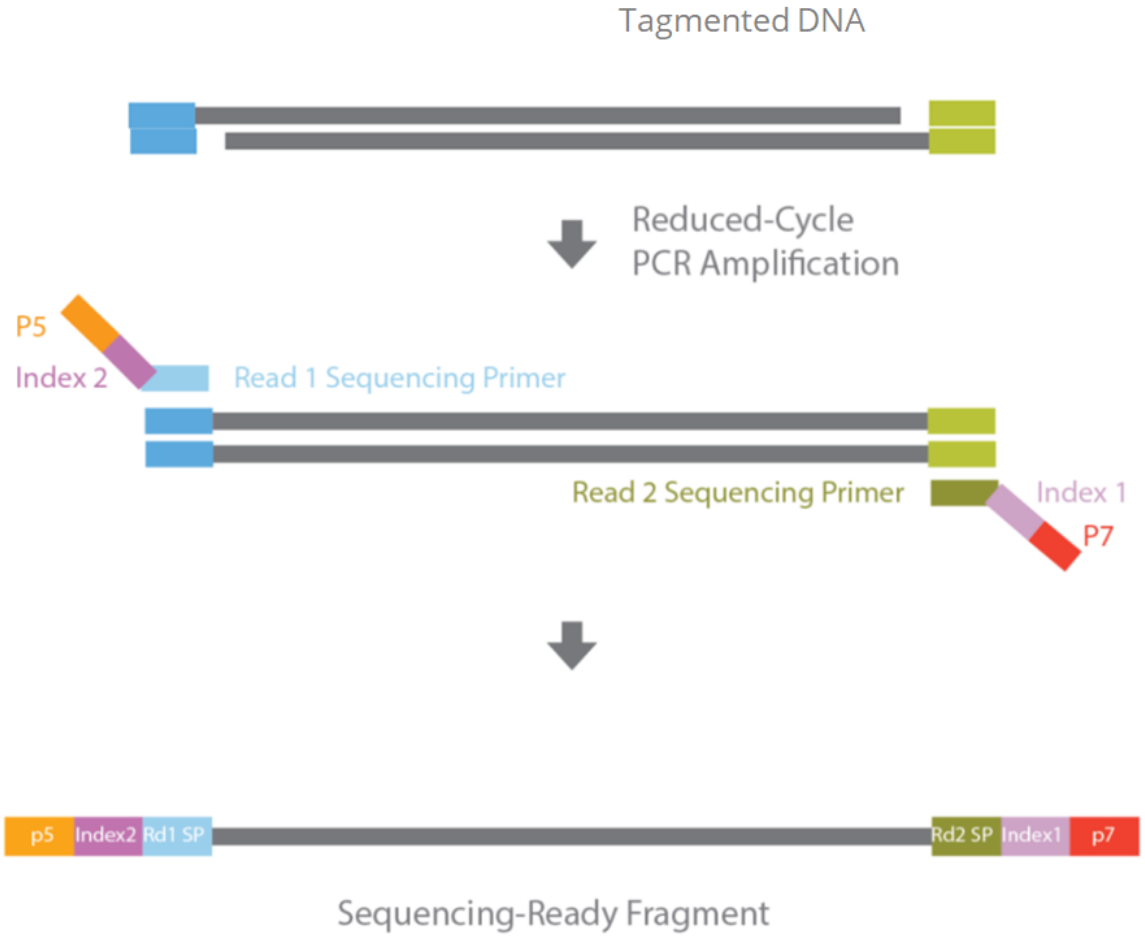


WES/Capture workflow

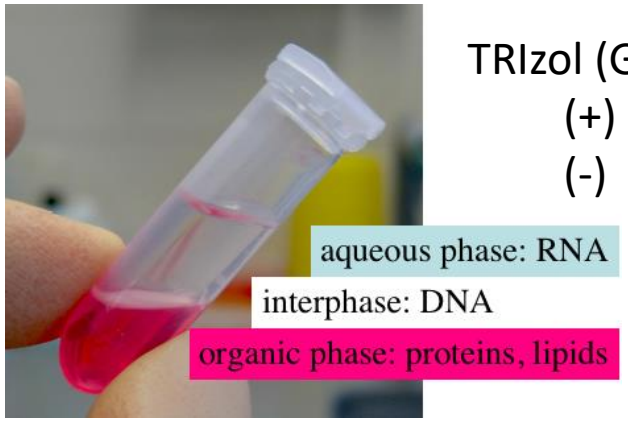


Tagmentation





RNA extraction



TRIzol (Guanidinium-acid-phenol)

(+) High yield/mg starting material, high quality

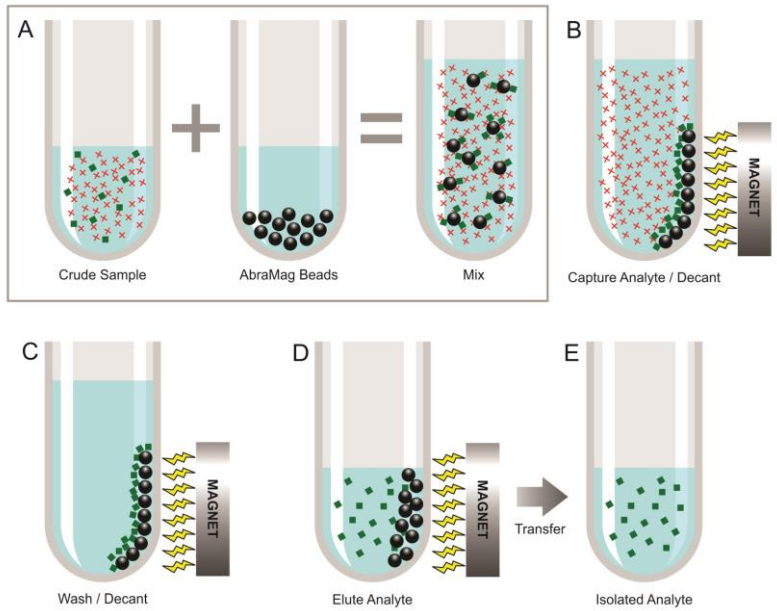
(-) Slower, hard to automate, organic solvents

Don't skip DNase step!

Magnetic particles

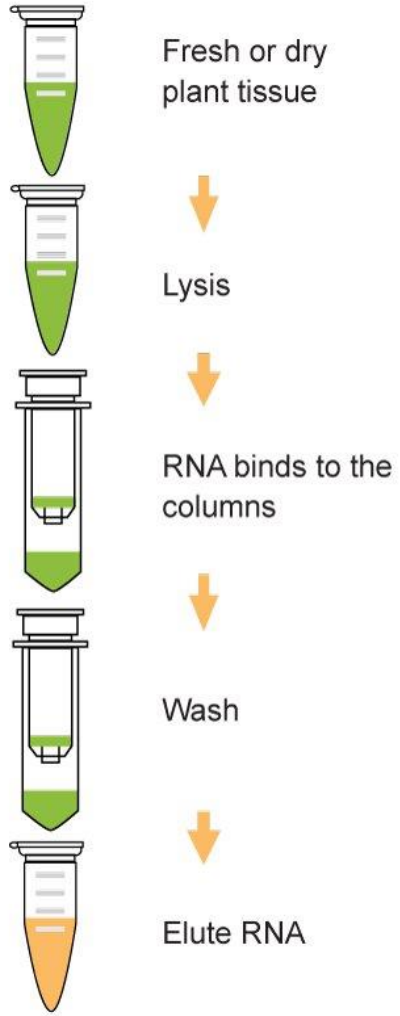
(+) Fast, easy to automate, efficient

(-) Carry-over of magnetic particles



Abraxis website

EpiGentek website



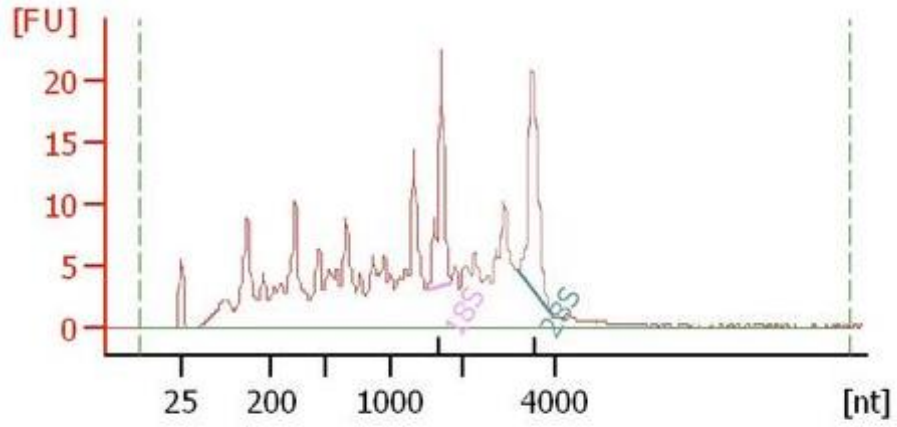
Column-based

(+) Convenient, can be automated

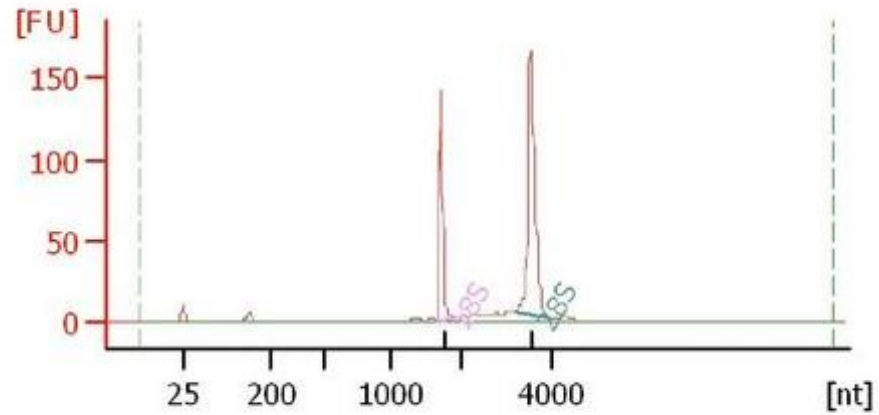
(-) Clogging, fixed capacity/yield, DNA carryover

RNA QC

- A260/280 ratio of ~2.0 (1.9-2.1 is acceptable)
- RNA integrity number (RIN) score: 0 (bad) to 10 (good)



RIN = 6.0



RIN = 10

- RIN values of >6 are sufficient for library prep
  - RIN<6 is still viable for library prep but harder to compare to RIN>6

RNA -> cDNA*

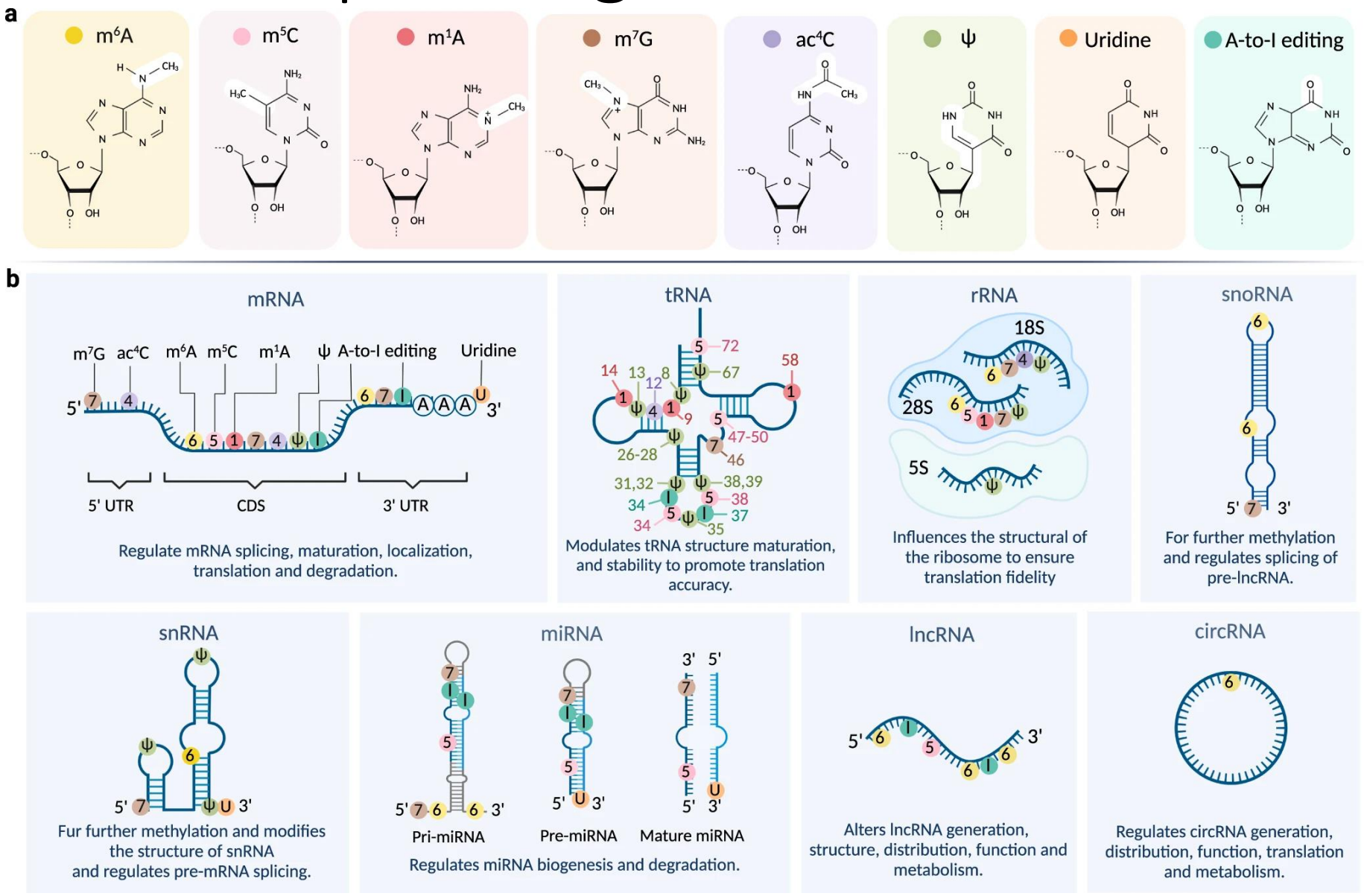
Start with total RNA

- 80% rRNA, 1-3% polyA+ RNA, tRNA, smallRNA (miRNA, snoRNA, piRNA), lincRNA

Strategy	Type of RNA	Unprocessed RNA content	Isolation method
Poly A+ selection	Coding	Low	Poly(dT) bead capture or primer
rRNA depletion	Coding, noncoding	High	Remove rRNA (hybridization or digestion)
Small RNA ligation	noncoding	High	smallRNA-specific adapter ligation

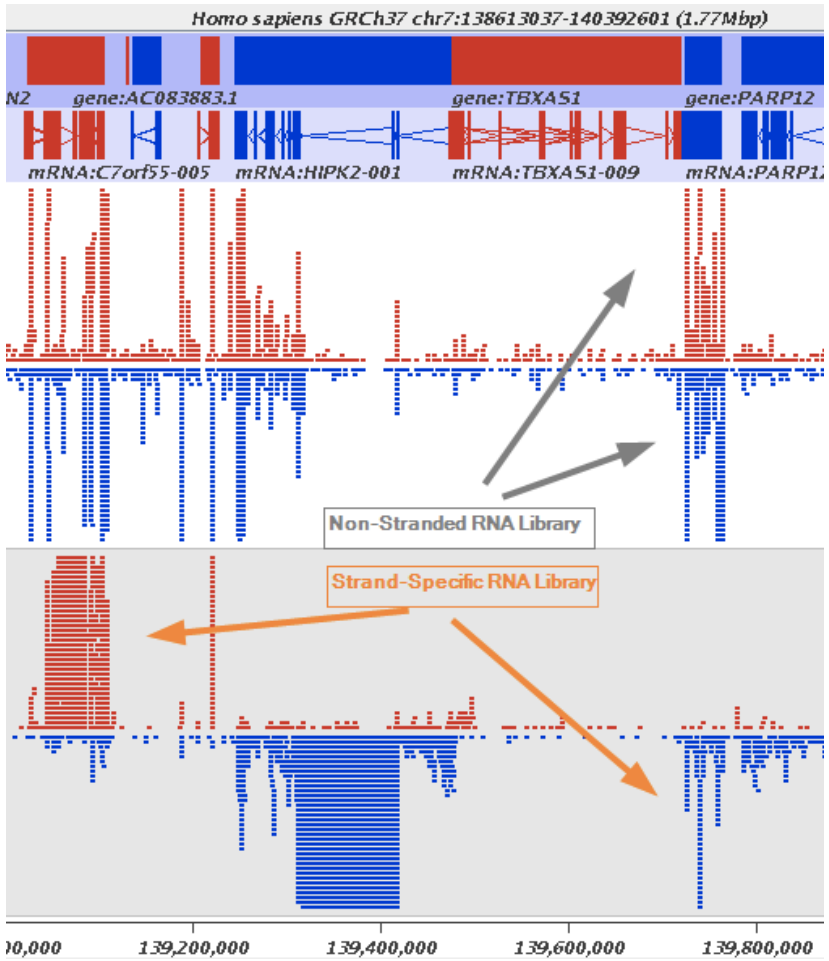
*N.B. Direct RNA sequencing is also possible with Oxford Nanopore

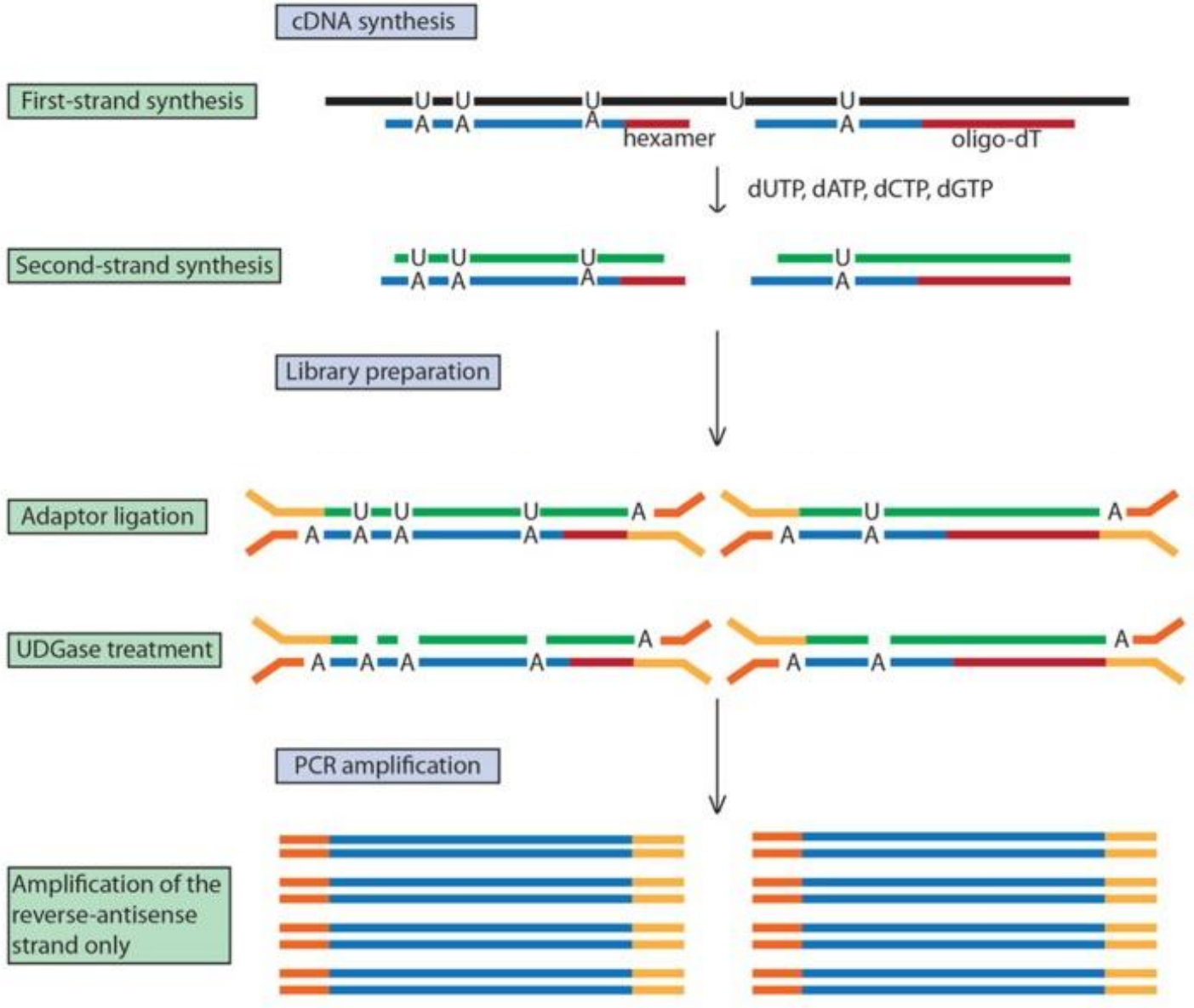
direct RNA sequencing-> modification



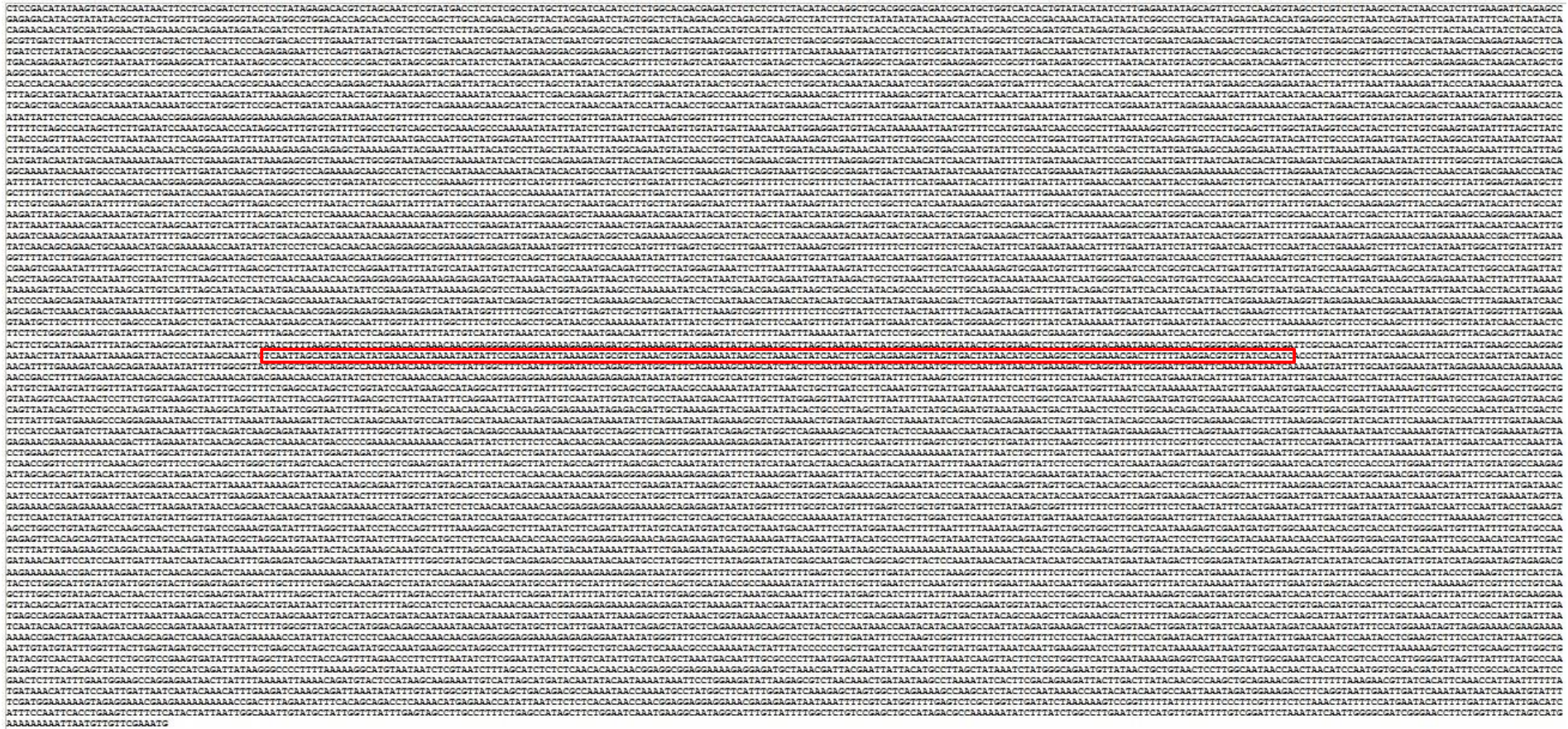
Signal Transduction and Targeted Therapy (*Sig Transduct Target Ther*) ISSN 2059-3635 (online)

Stranded RNA-seq - Illumina



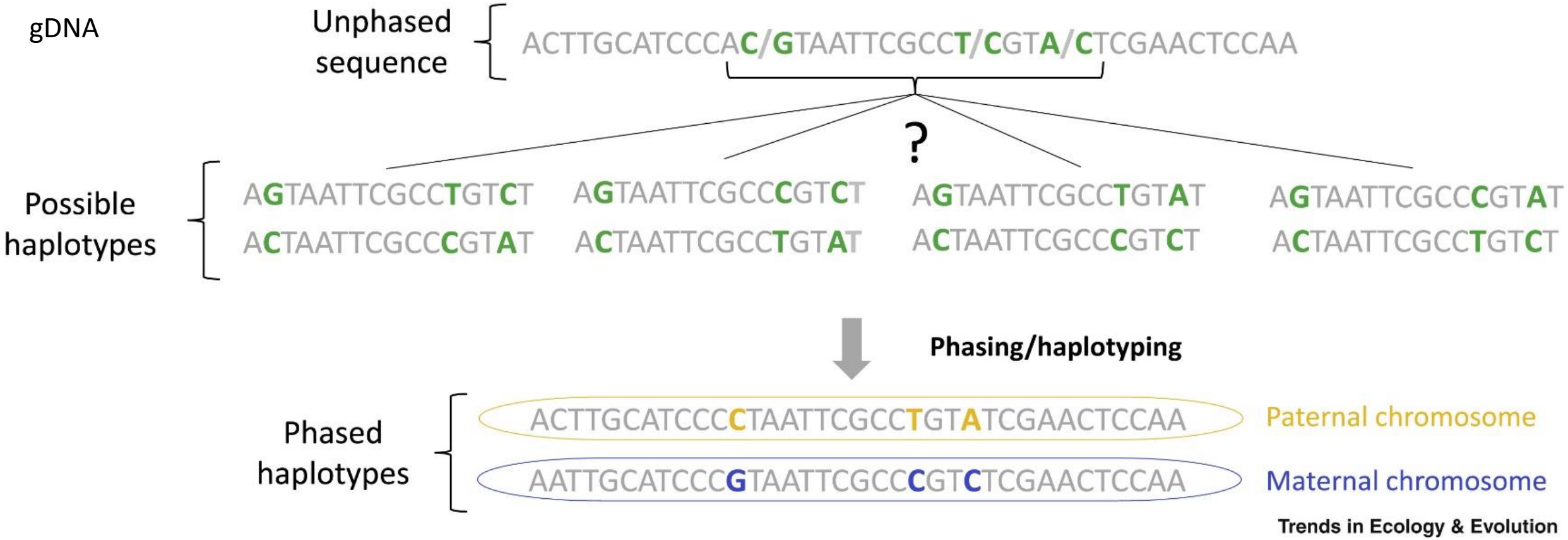


Long Read vs Short Read



> 22Kb Aligned Read

Long Read – Enriches detection in gDNA



Long Read – Enhances detection in cDNA

Transcriptome

↓

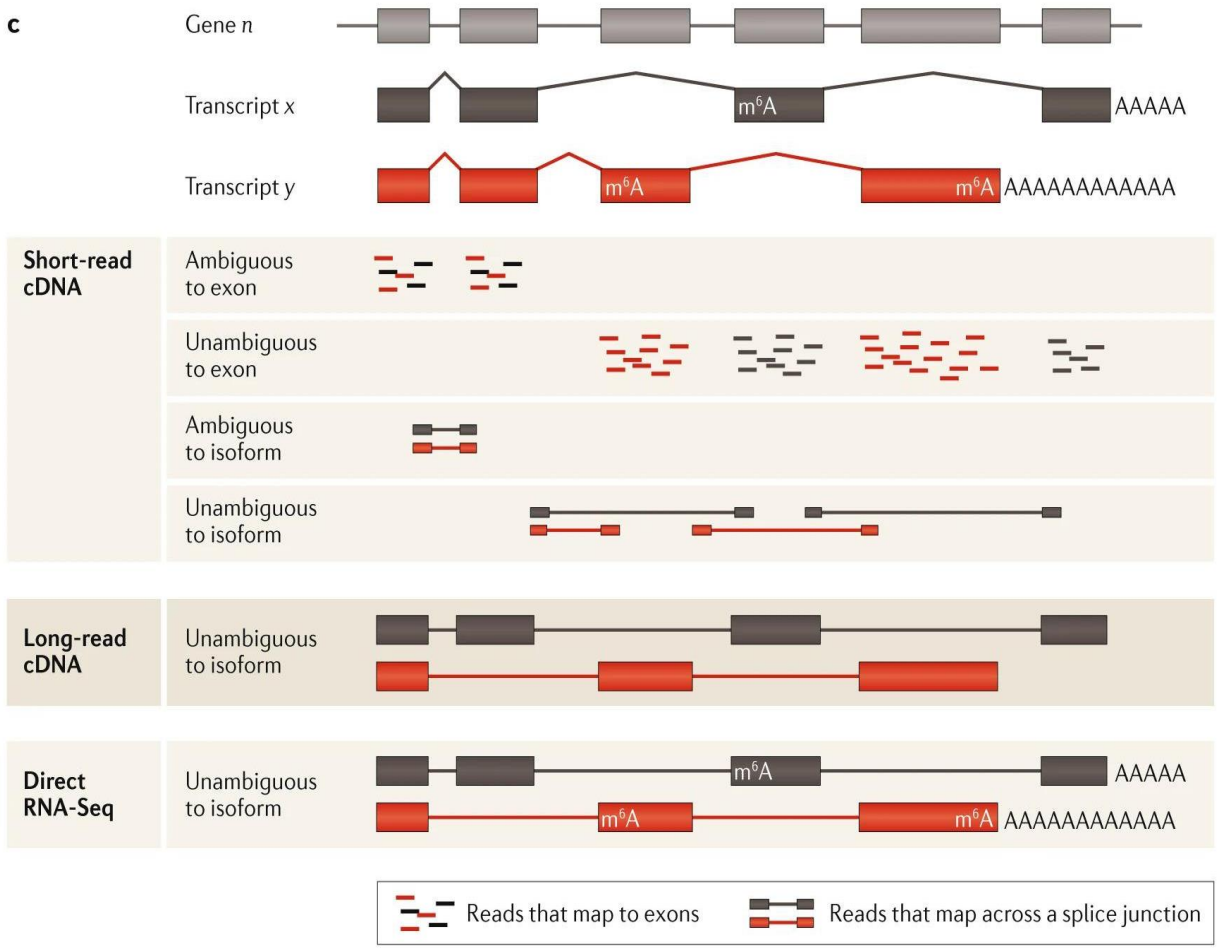
Illumina

↓

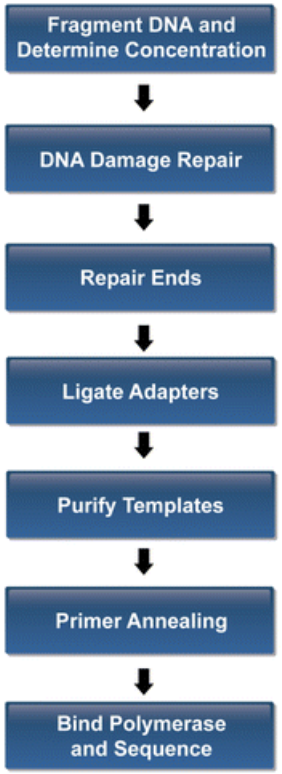
PacBio/ONT

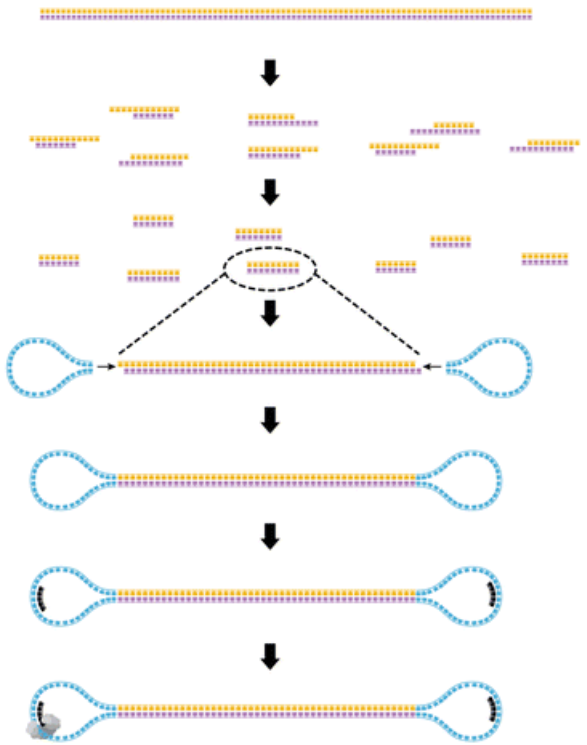
↓

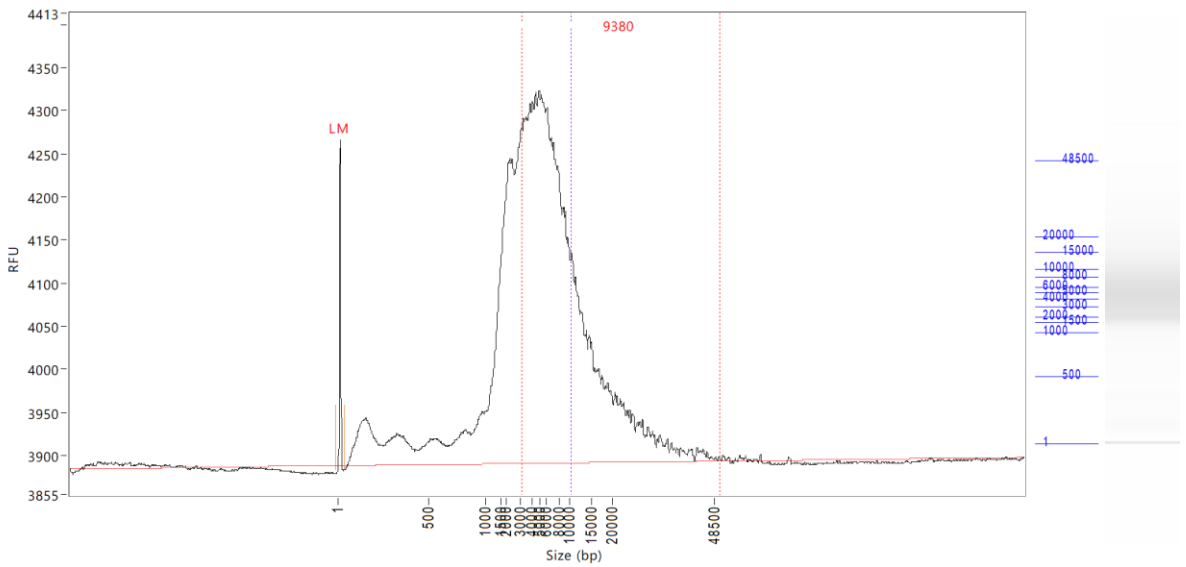
ONT



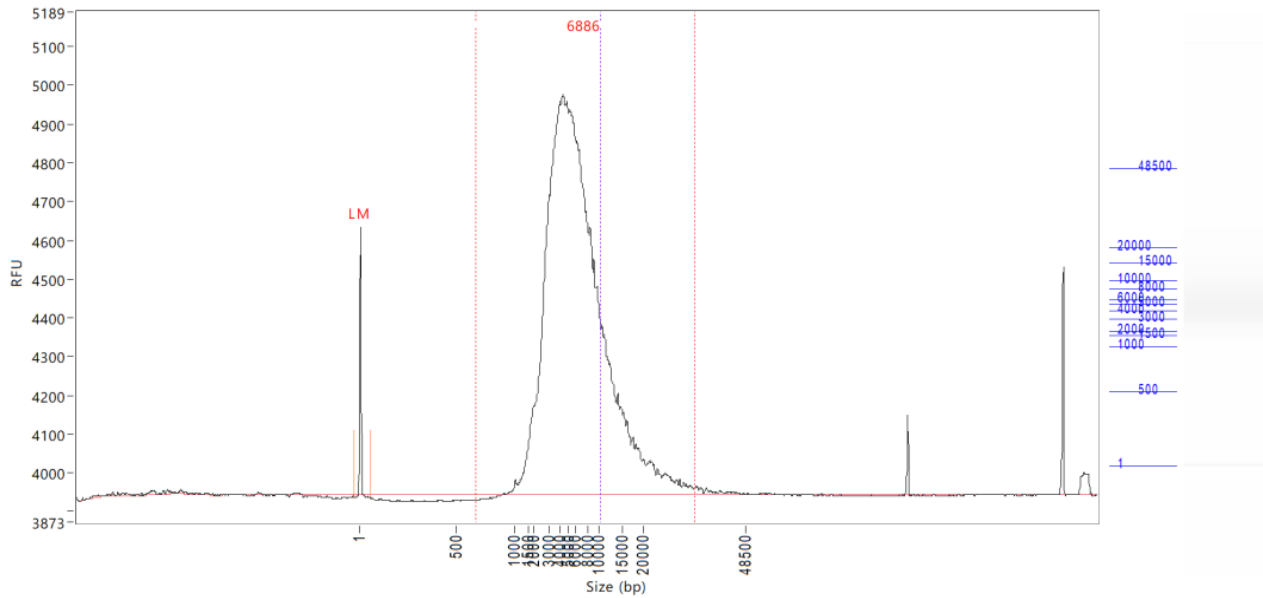
Sample prep - PacBio





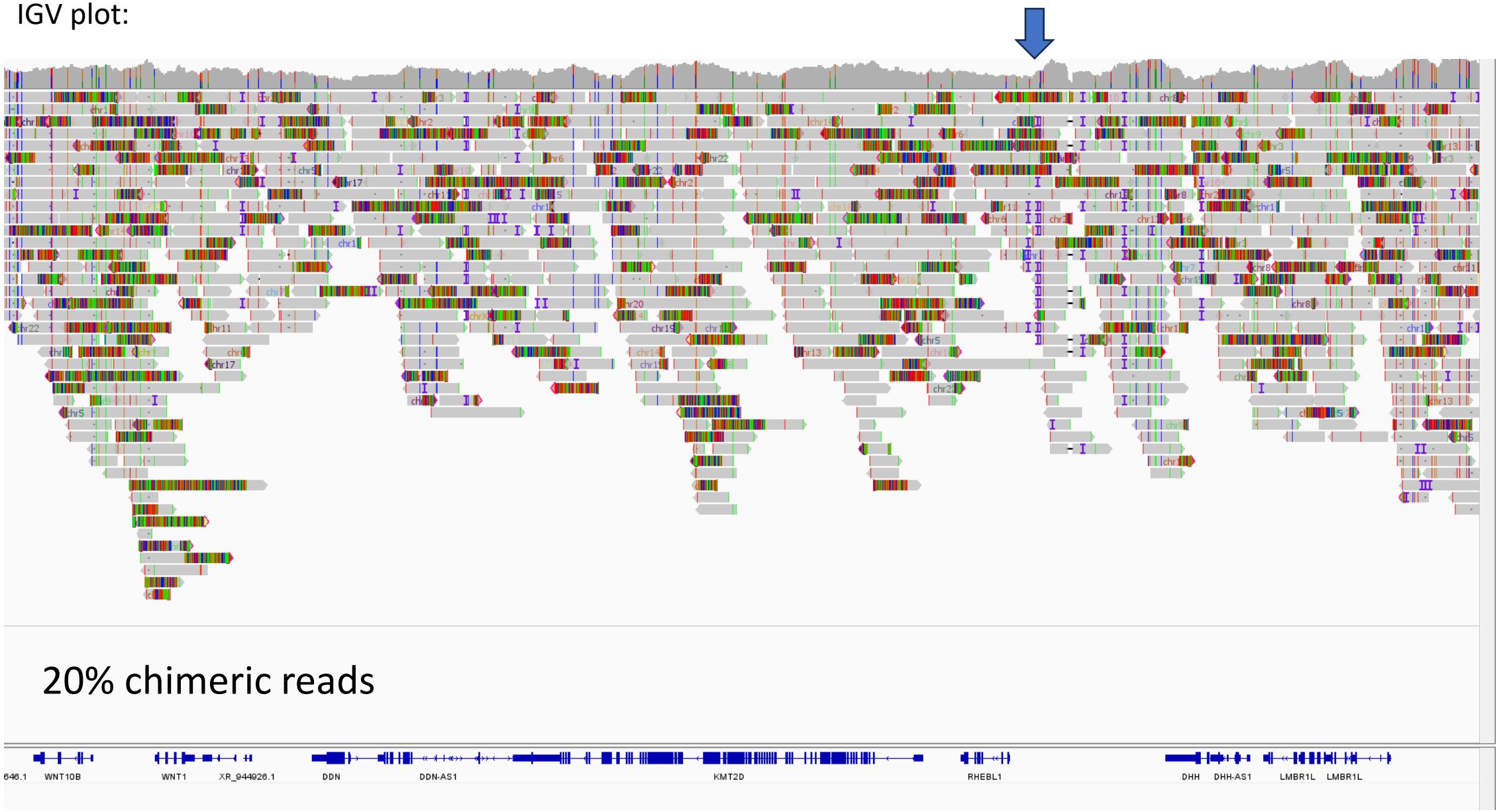


Size Selection followed by Standard Library prep PacBIo



Sample prep – PacBio chimeric reads

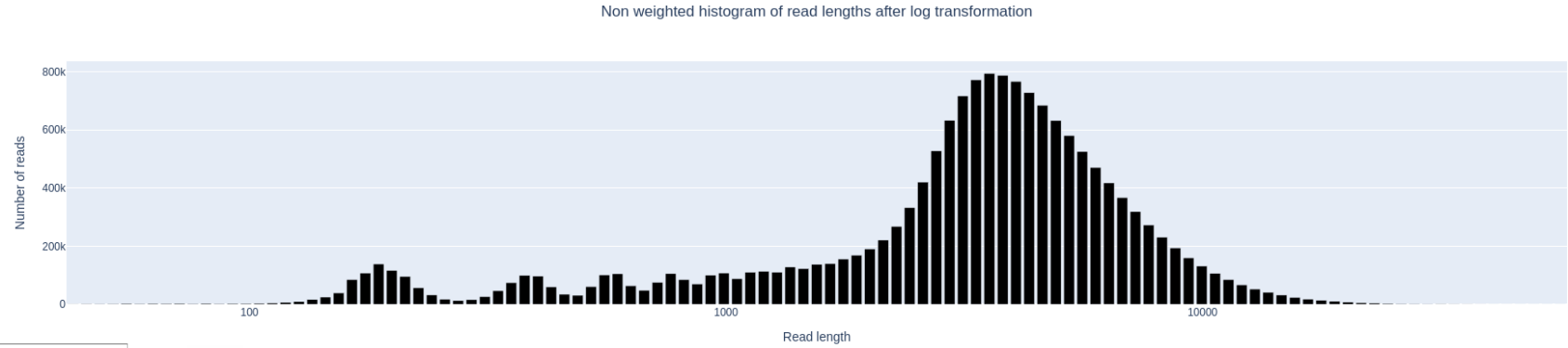
IGV plot:

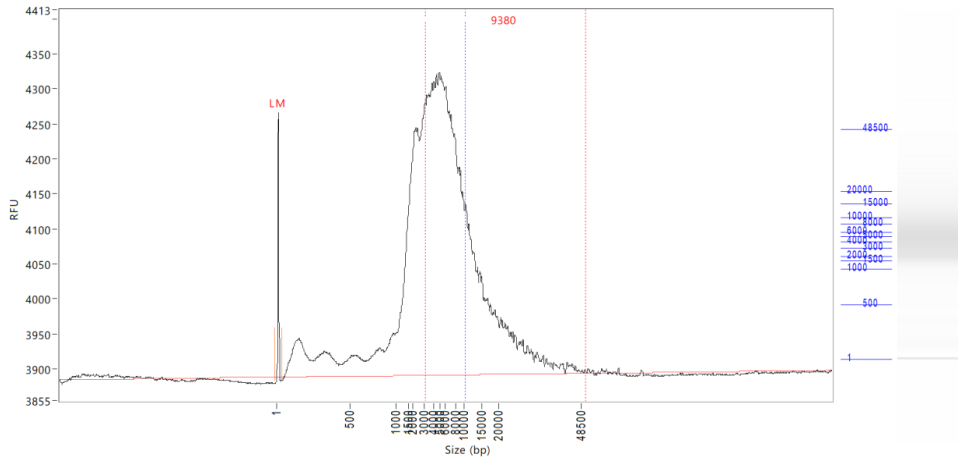


20% chimeric reads

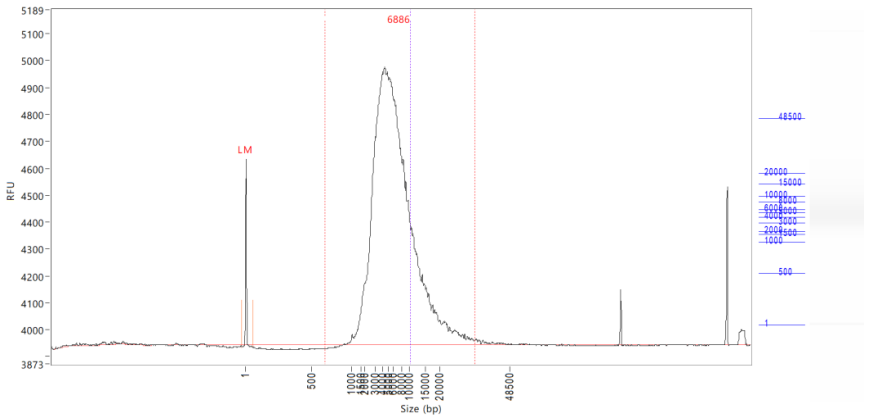
Sample prep artefacts – PacBio chimeric reads

Insert-Size Mapped reads:

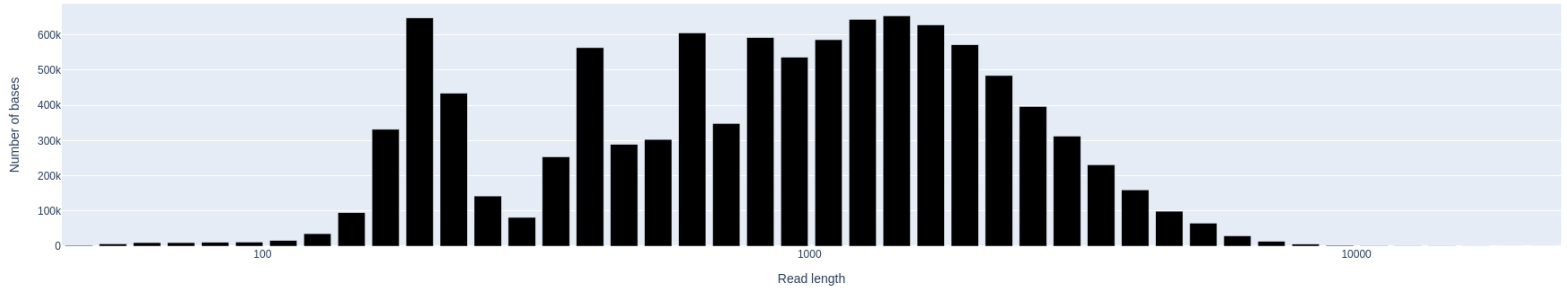




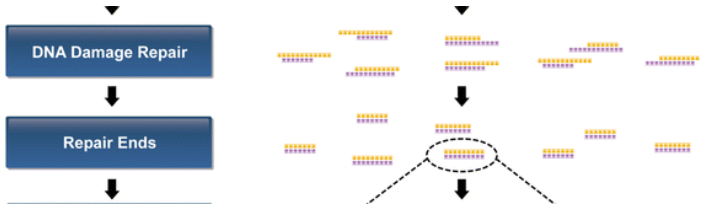
size selection



Library prep

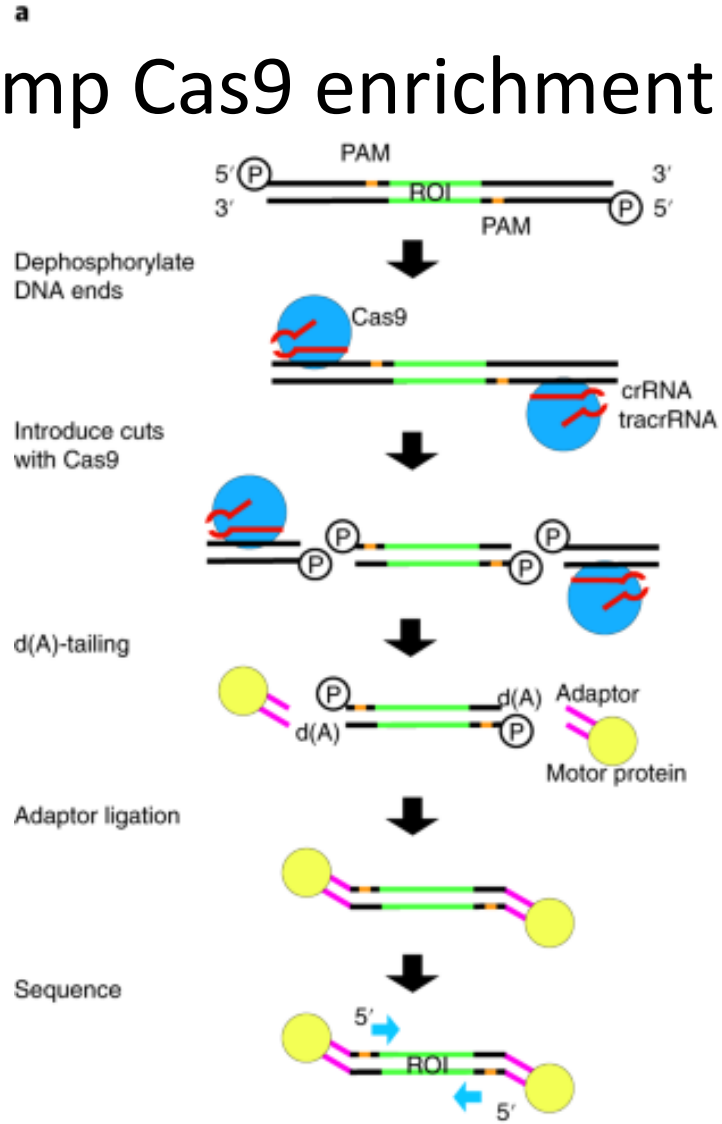


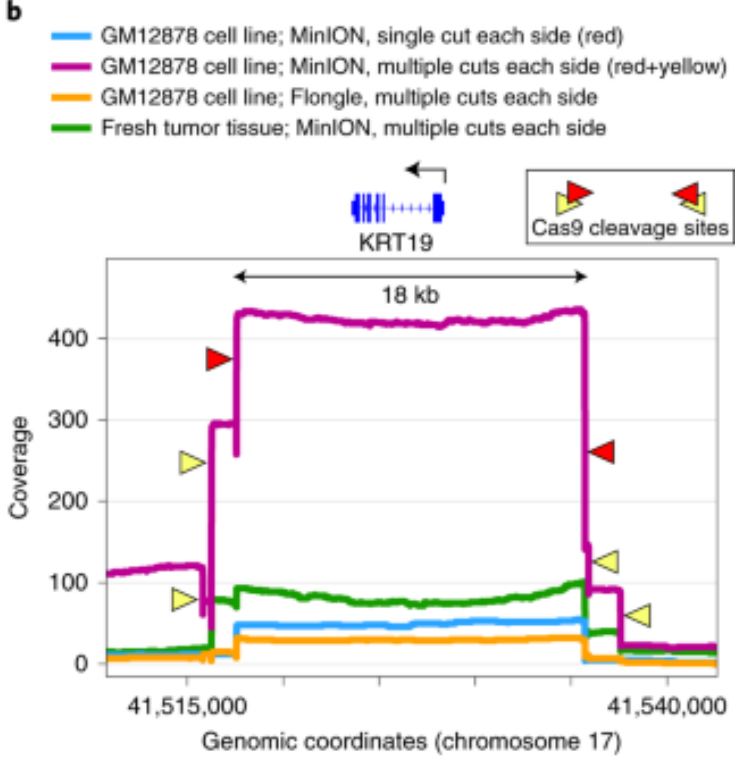
Insert-Size soft clipped reads:



Oxford Nanopore: NoAmp Cas9 enrichment

- PacBio kit is called **PureTarget**
- High Depth >200x
- Native gDNA sequenced
- DNA modification retained
- No Amplification bias
- Multiplexing Samples





c

10 sites (12–24 kb)

	GM12878			Breast tumor
	One cut each side	Multiple cuts (two or three) each side		
	MinION	MinION	Flongle	minION
Aligned reads	75,822	235,328	37,715	611,992
On-target reads	1,370	10,854	790	2,702
On-target percentage	1.81	4.61	2.09	0.44
Median coverage	80	680	34	93

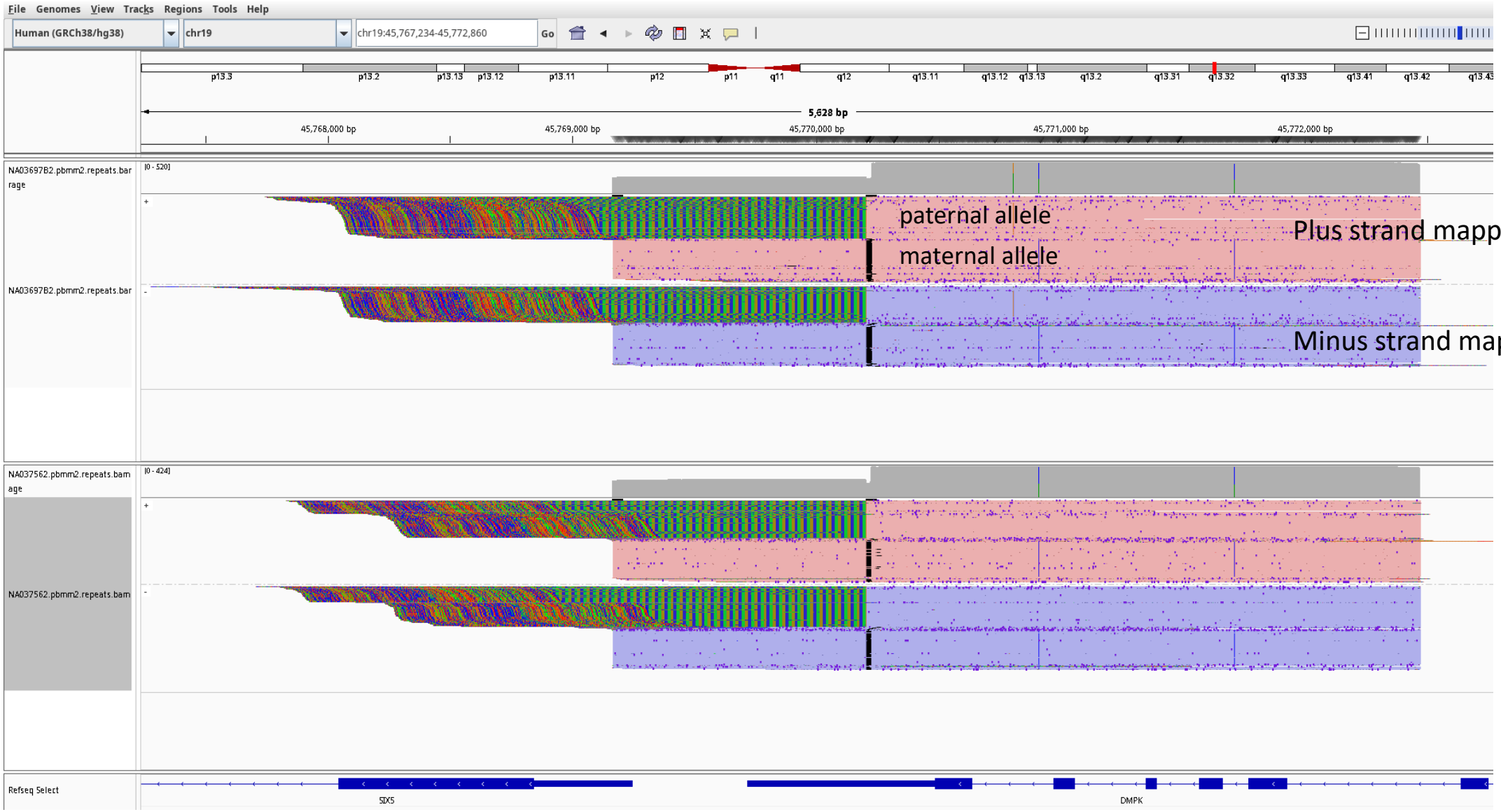
Targeted nanopore sequencing with Cas9-guided adapter ligation. *Nat Biotechnol* **38**, 433–438 (2020). https://doi.org/10.1038/s41587-020-0407-5

PacBio PureTarget demo data - Sequellle:

https://downloads.pacbcloud.com/public/dataset/PureTargetRE/

Sample 1

Sample 2

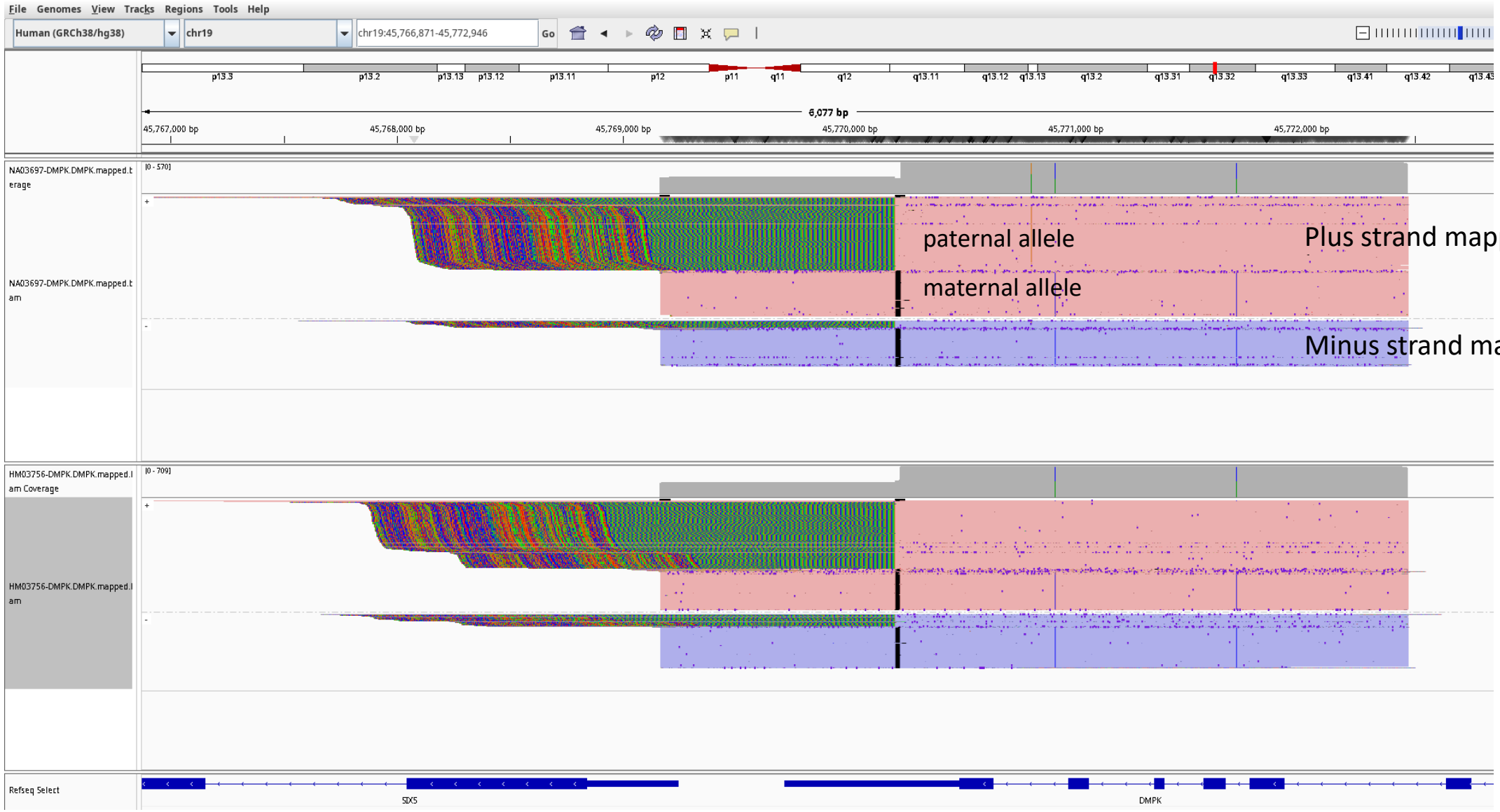


PacBio PureTarget demo data - **Revio - same samples**:

https://downloads.pacbcloud.com/public/dataset/PureTargetRE/

Sample 1

Sample 2

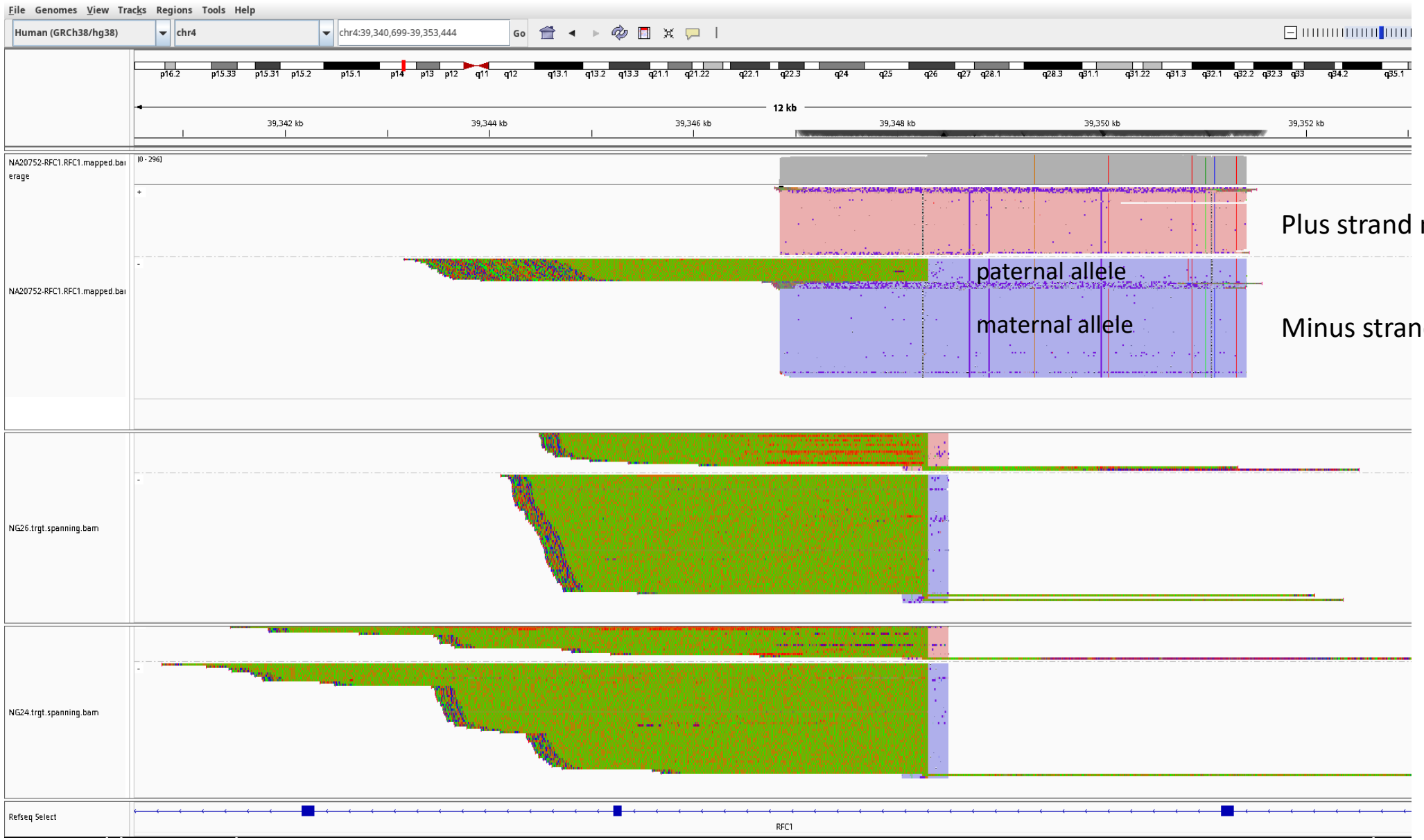


PacBio PureTarget demo data - Revio:

Different samples, different target

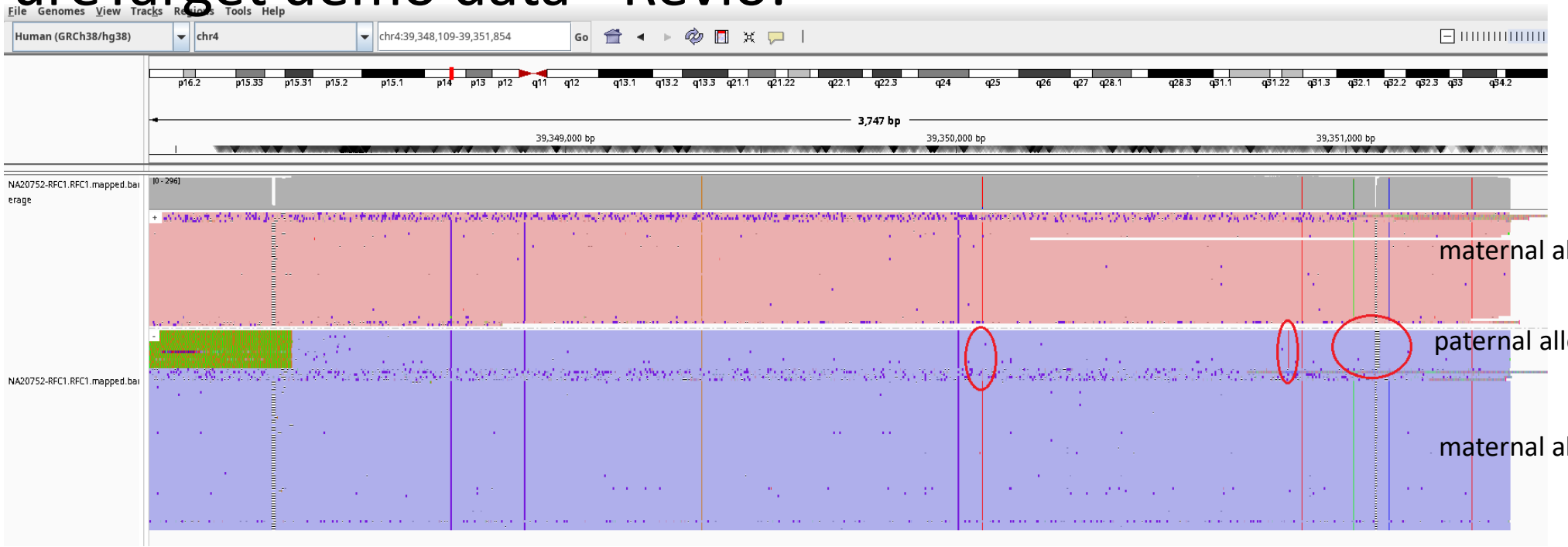
Sample 3

Sample 4

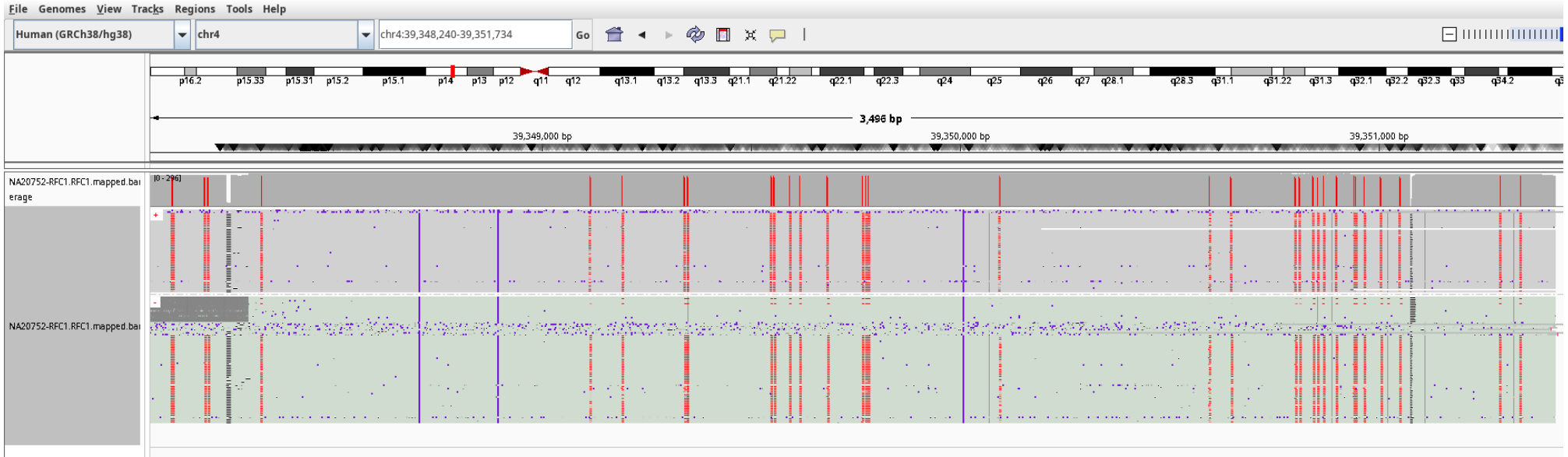


PacBio PureTarget demo data - Revio:

Sample 3



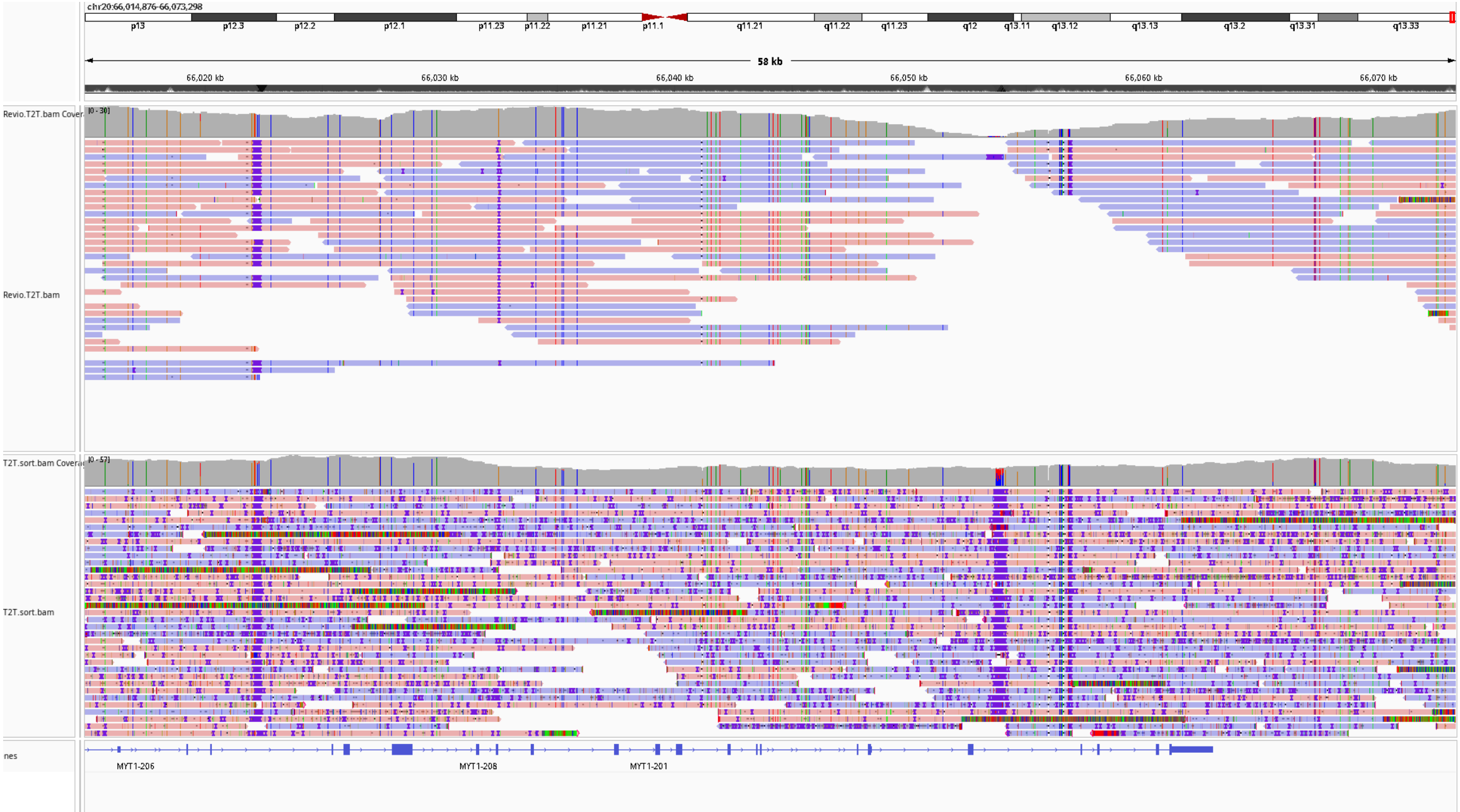
Sample 3: Methylation Profile (5mC)



Benchmark ONT and PacBio Revio WGS (40x)

Revio

ONT



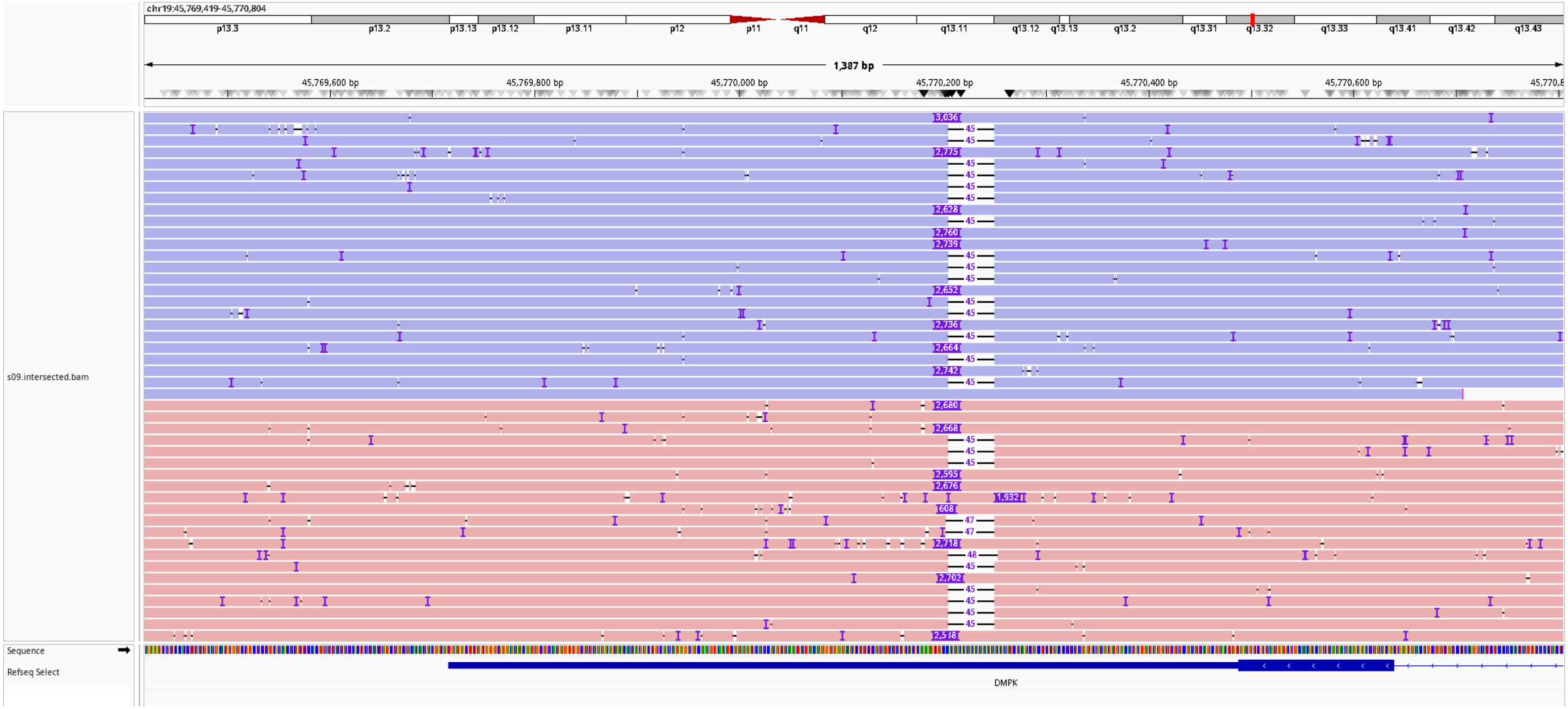
ONT and PacBio Revio Benchmark (2)

Revio

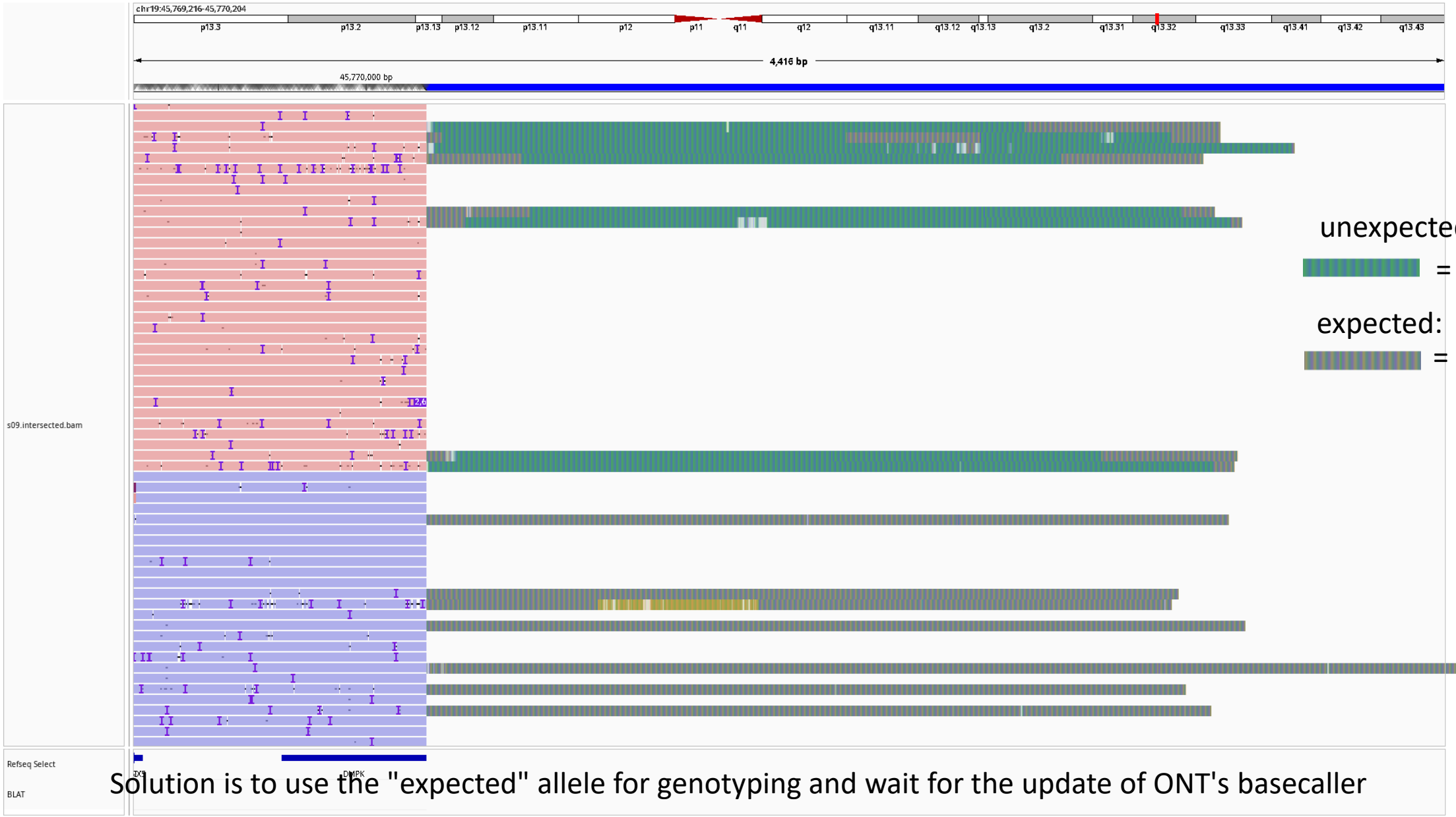
ONT



Cas9 enrichment ONT



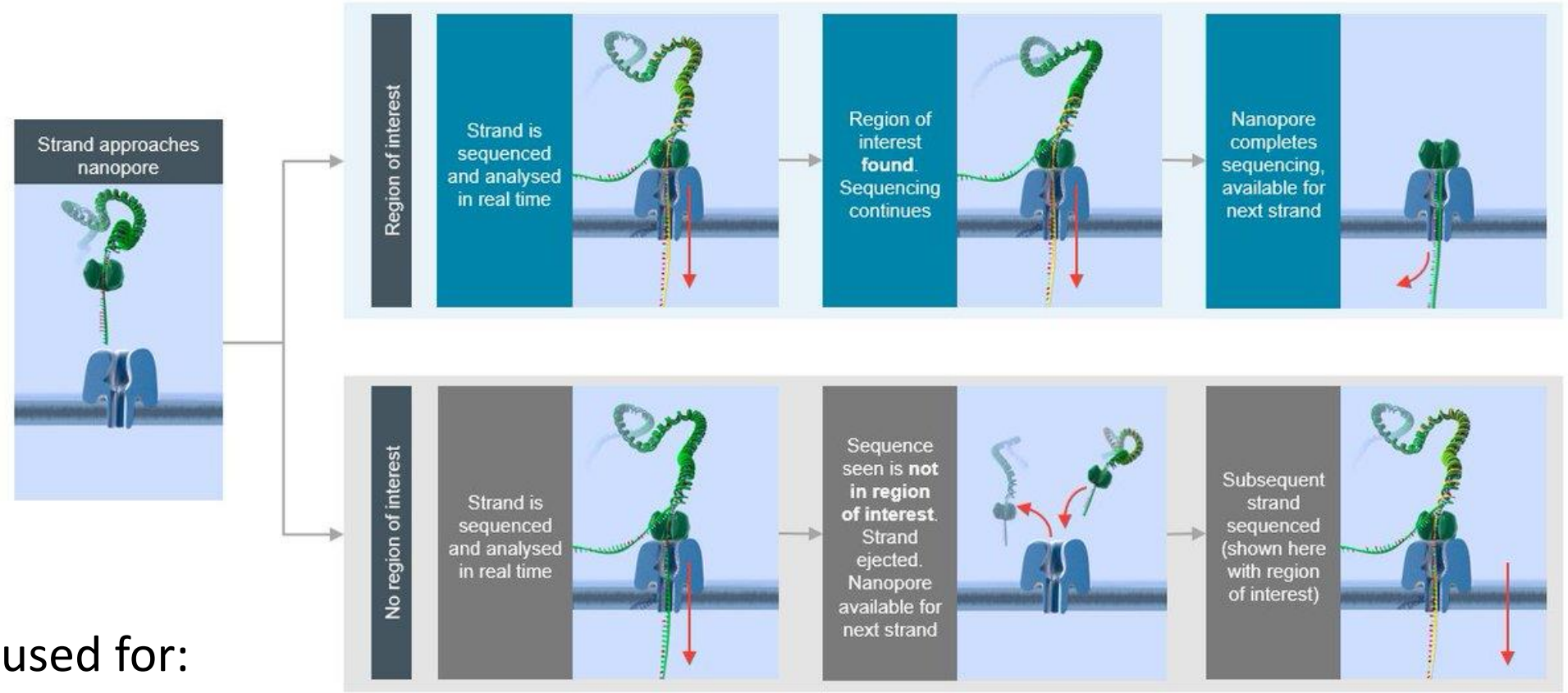
Cas9 enrichment ONT – discrepancy between basecalls



Solution is to use the "expected" allele for genotyping and wait for the update of ONT's basecaller

Oxford Nanopore: Adaptive Sampling

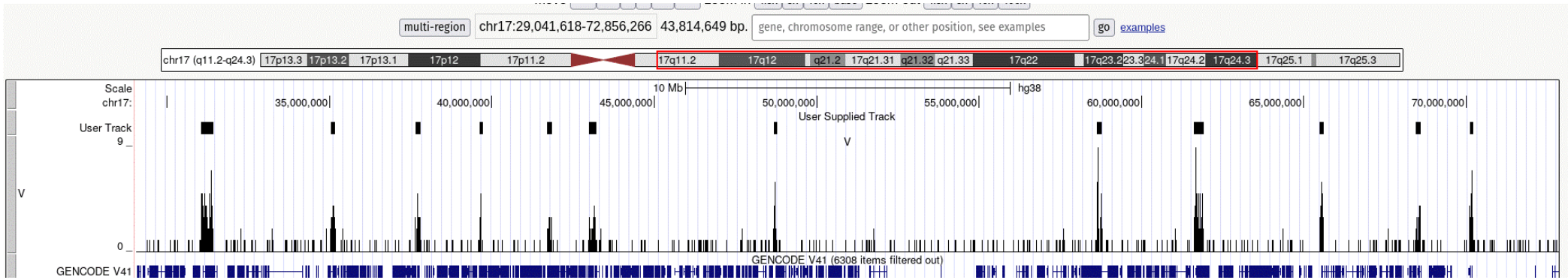
"Adaptive Sampling" is Selective sequencing of targeted regions



Adaptive Sampling used for:

- gDNA (including methylation)
- Amplicon Balancing
- Barcode Balancing

Adaptive Sampling requirements
Effect of 3h running:

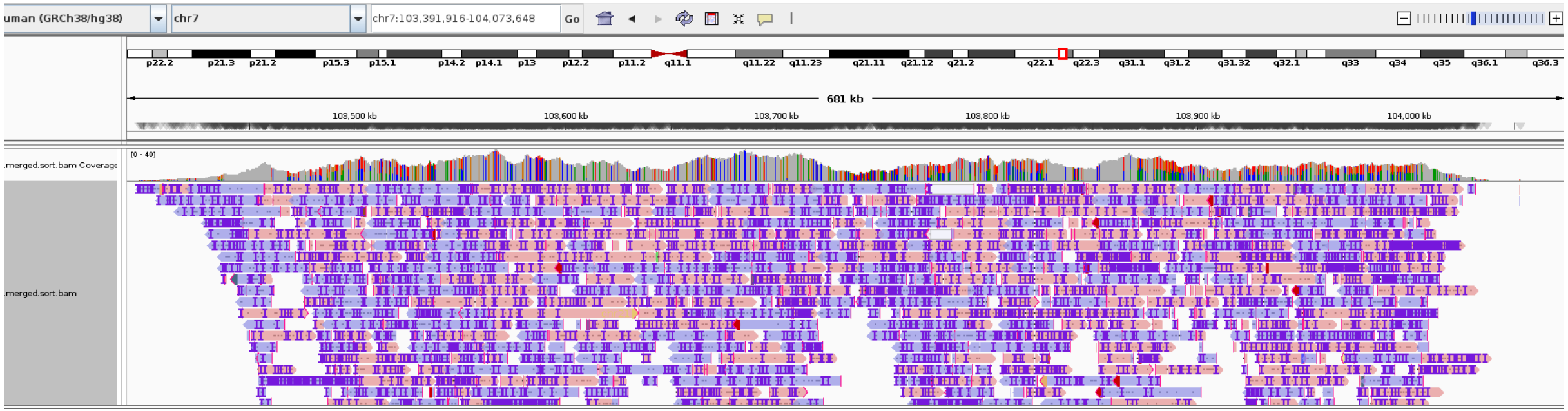



```
Target list:
chr17          17162212        17287188        +
chr17          17162212        17287188        -
chr17          31044927        31432116        +
chr17          31044927        31432116        -
chr17          35042208        35171522        +
chr17          35042208        35171522        -
chr17          37636431        37795105        +
chr17          37636431        37795105        -
chr17          39615349        39716554        +
chr17          39615349        39716554        -
chr17          41704604        41836931        +
chr17          41704604        41836931        -
chr17          42994295        43220245        +
chr17          42994295        43220245        -
chr17          48674763        48778750        +
chr17          48674763        48778750        -
chr17          58642573        58785611        +
chr17          58642573        58785611        -
chr17          61629139        61913528        +
chr17          61629139        61913528        -
chr17          65478563        65611648        +
chr17          65478563        65611648        -
```

Easy setup:

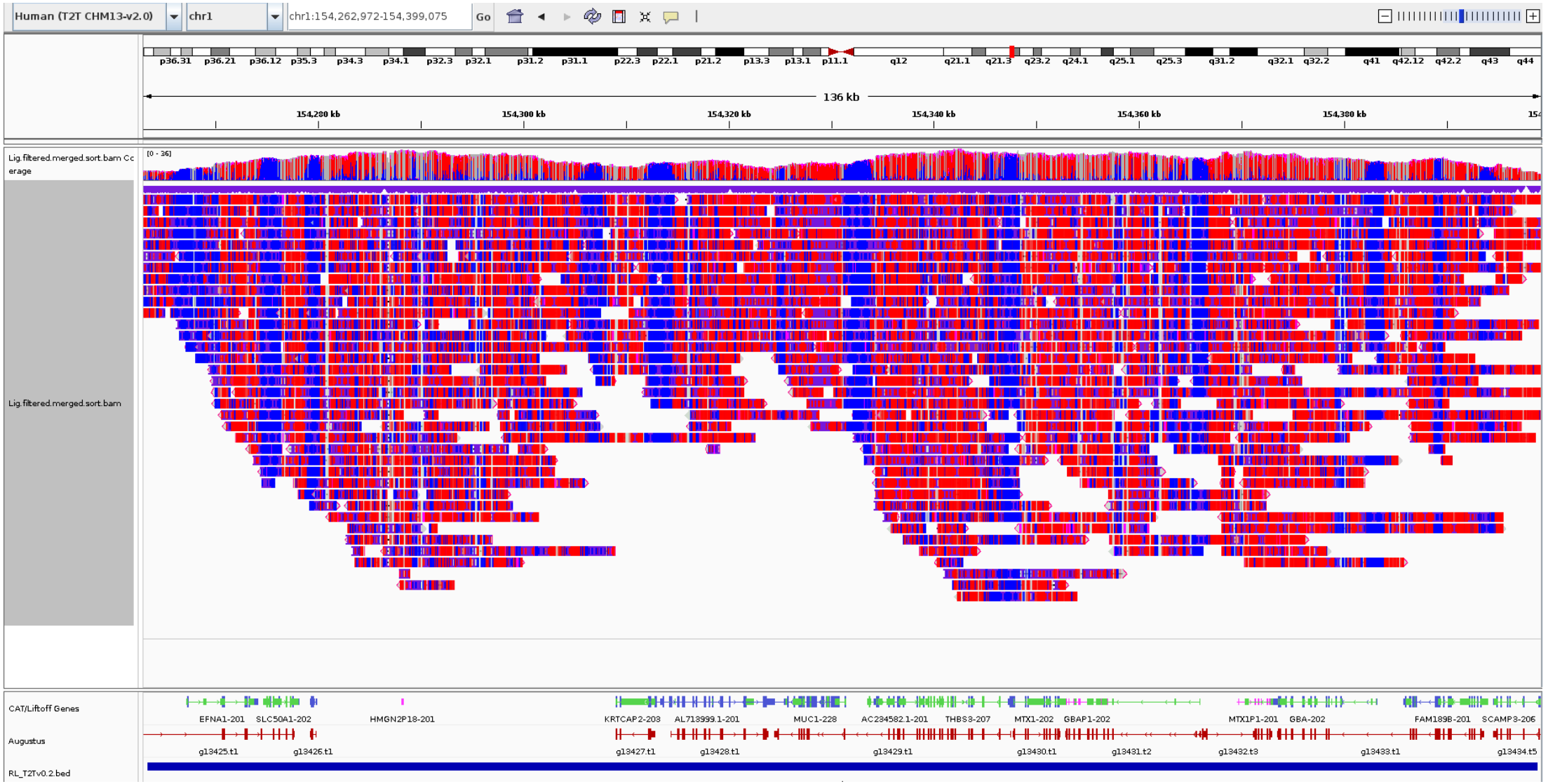
- High end GPU
- Requirement: BED file
- Maximum size of region of interest: 300Mb
- Minimum size of region of interest: 300kb

Adaptive Sampling – Full Run

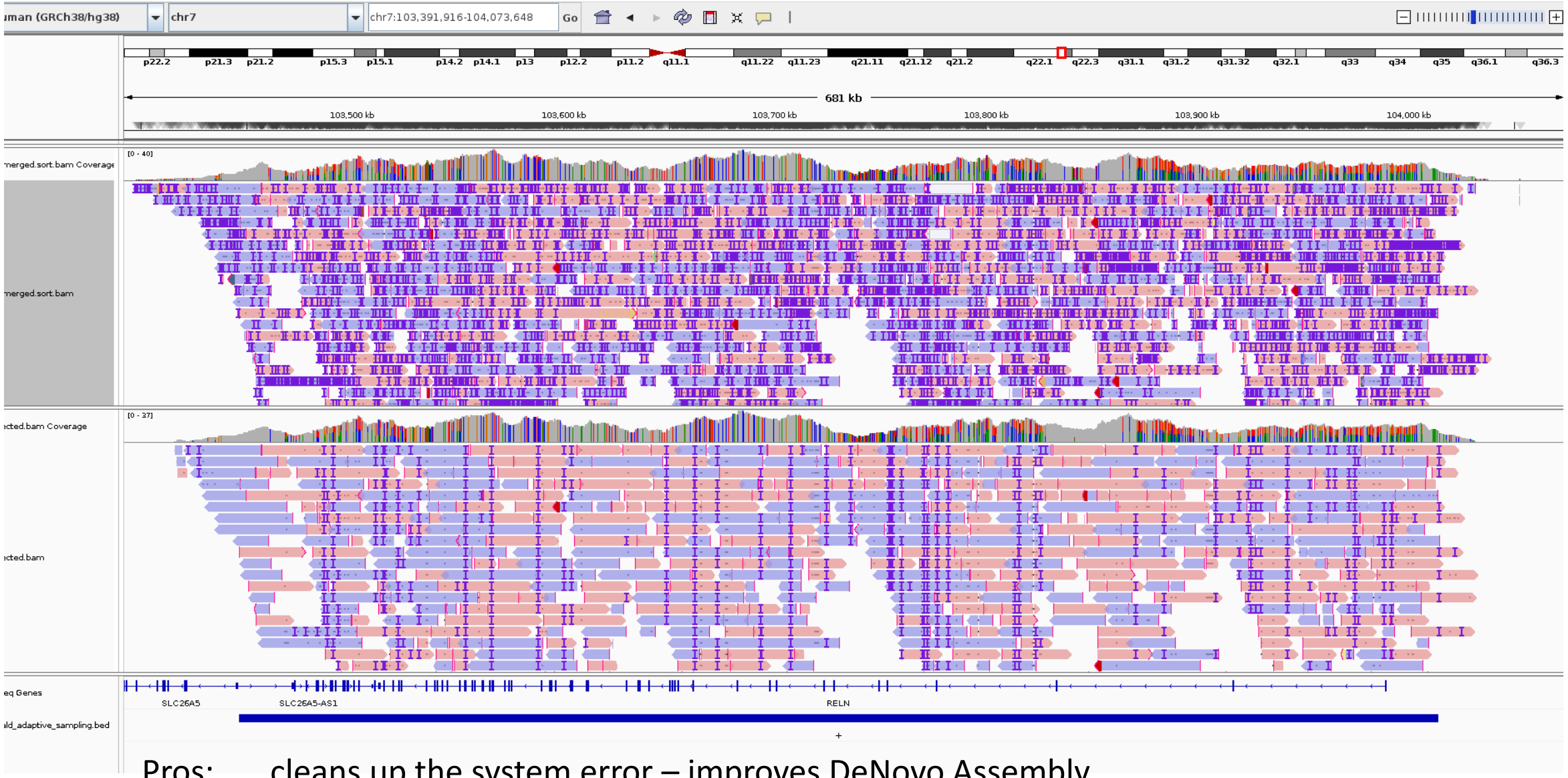


- Multiplexing 4-6 sample per PromethION chip
- Run for 72h
- Enough depth for Germline SNV/SV detection

Adaptive Sampling – DNA Modifications are retained



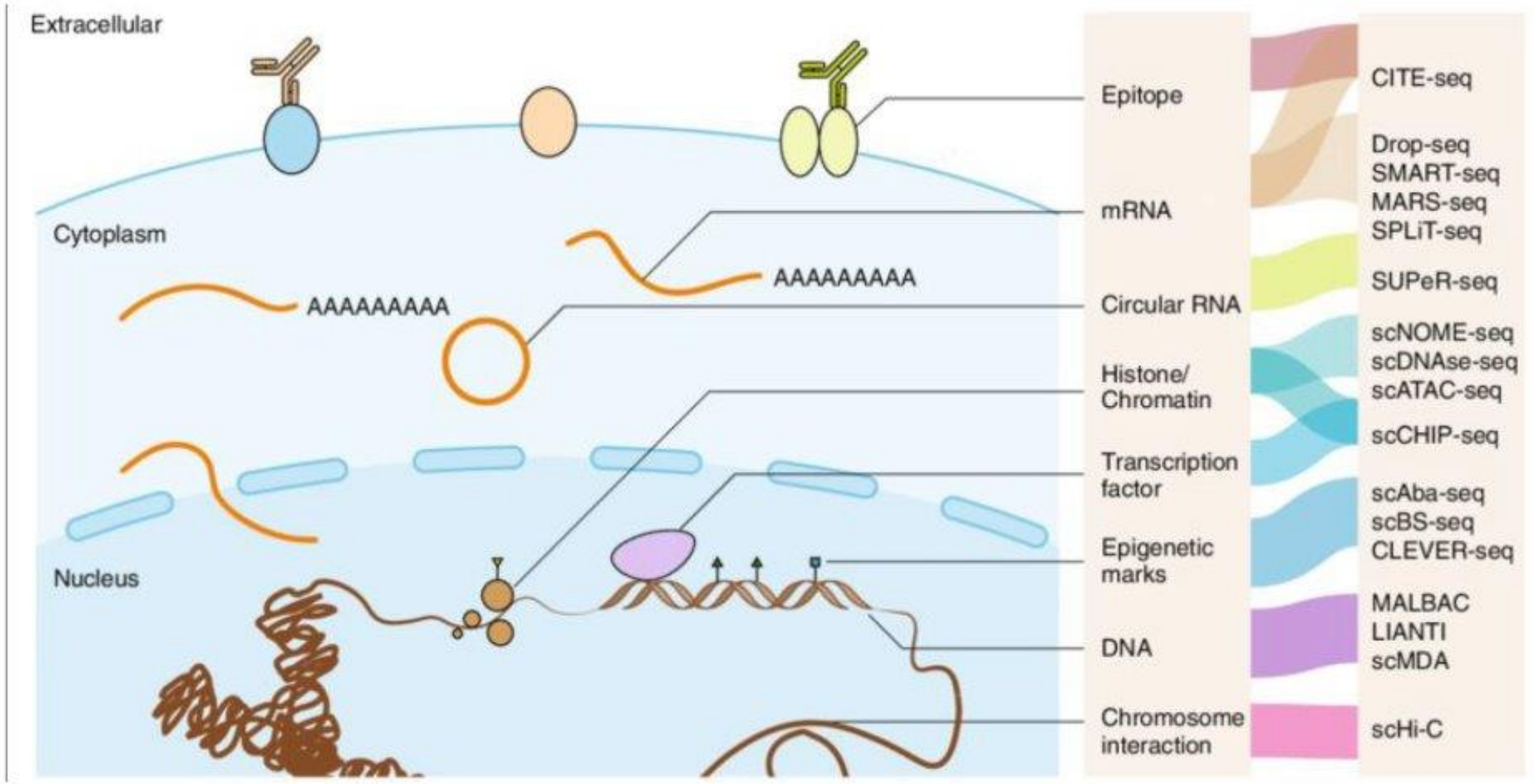
Dorado correct – remove noise from ONT data



Pros: cleans up the system error – improves DeNovo Assembly

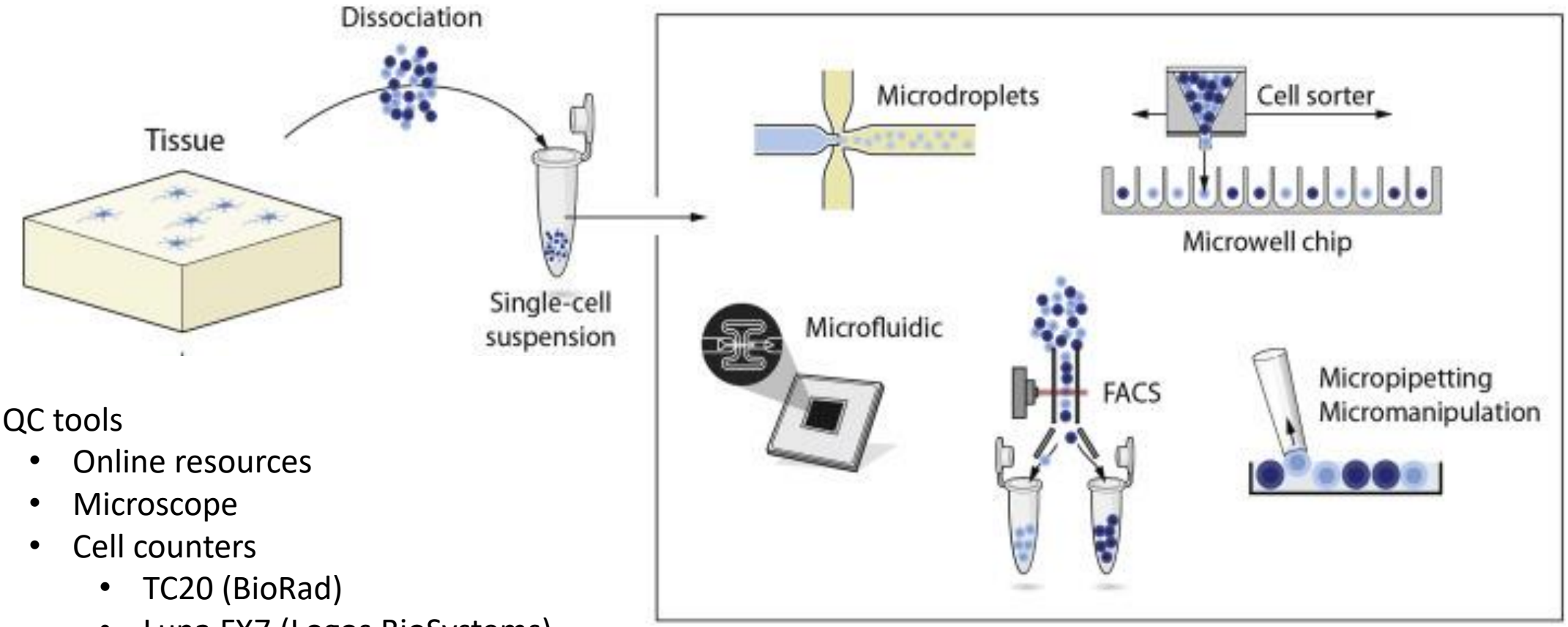
Cons: requires >10kb fragments, GPU

Single-cell omics



Understanding tumor ecosystems by single-cell sequencing: promises and limitations. Genome Biology. 19. 10.1186/s13059-018-1593-z.

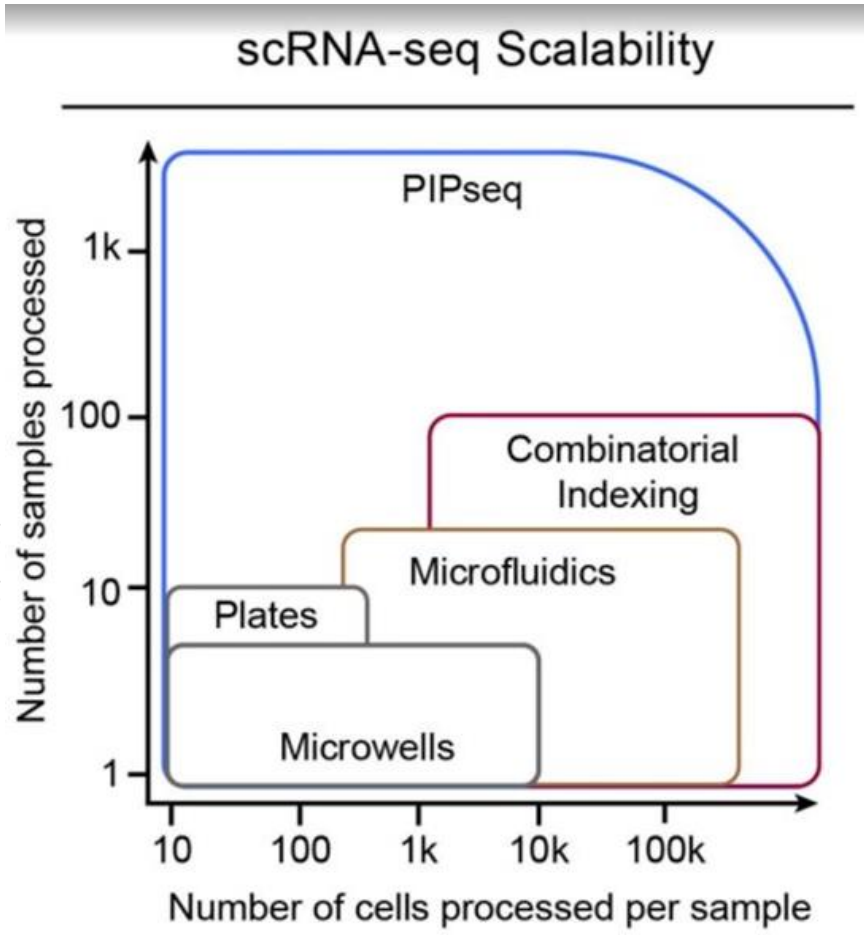
Single-cell sequencing – different ways



- QC tools
  - Online resources
  - Microscope
  - Cell counters
    - TC20 (BioRad)
    - Luna FX7 (Logos BioSystems)

Commercial Single-cell sequencing platforms

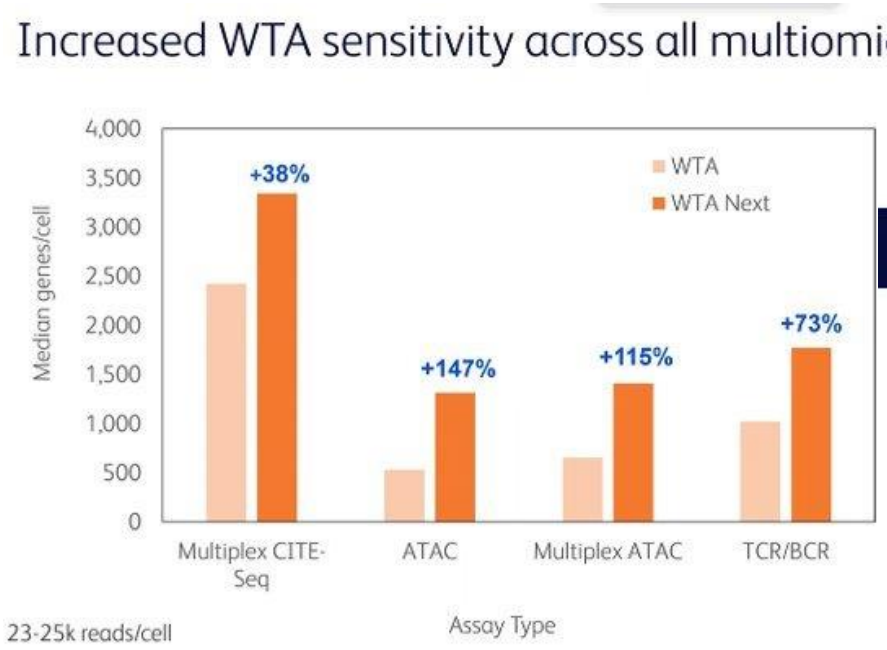
Kit Name	Abbreviation	Technology Group
Chromium Single Cell 3' Reagent Kit, v3.1 Chemistry. (10x Genomics)	10X 3'	Emulsion
Chromium Single Cell 5' Reagent Kit, v2 Chemistry. (10x Genomics)	10X 5'	Emulsion
Chromium Fixed RNA Profiling Reagent Kit. (10x Genomics)	10X FRP	Emulsion
PIPseq T20 3' Single Cell RNA Kit v4.0. (Fluent Biosciences)	Fluent	Emulsion
BD Rhapsody WTA Reagent Kit. (Becton Dickinson)	BD	Microwell
HIVE CLX SCRNA Seq Kit. (Honeycomb Biotechn.)	Honeycomb	Microwell
GEXSCOPE® Single Cell RNA Library Kit. (Singleron Biotech.)	Singleron	Microwell
Evercode WT v2. (Parse Biosciences)	Parse	Combinatorial Indexing
Single Cell RNA Kit. (Scale Biosciences)	Scale	Combinatorial Indexing
ASTERIA Single-cell RNASeq Kit. (Scipio Bioscience)	Scipio	Matrigel

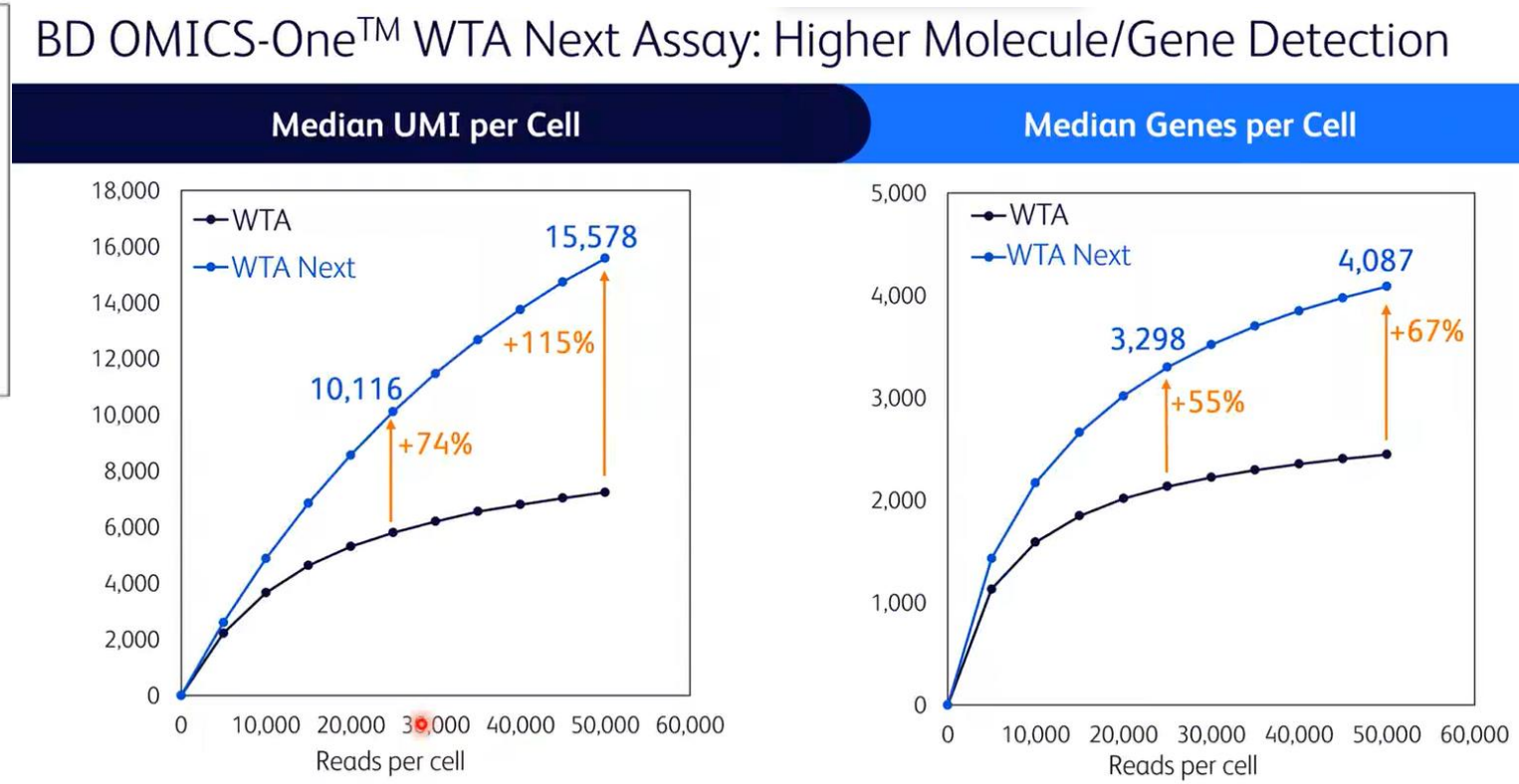


Comparative Analysis of Commercial Single-Cell RNA Sequencing Technologies: https://doi.org/10.1101/2024.06.18.599579

Single-cell sequencing – example of improvements

Check version numbers when reading and comparing performance



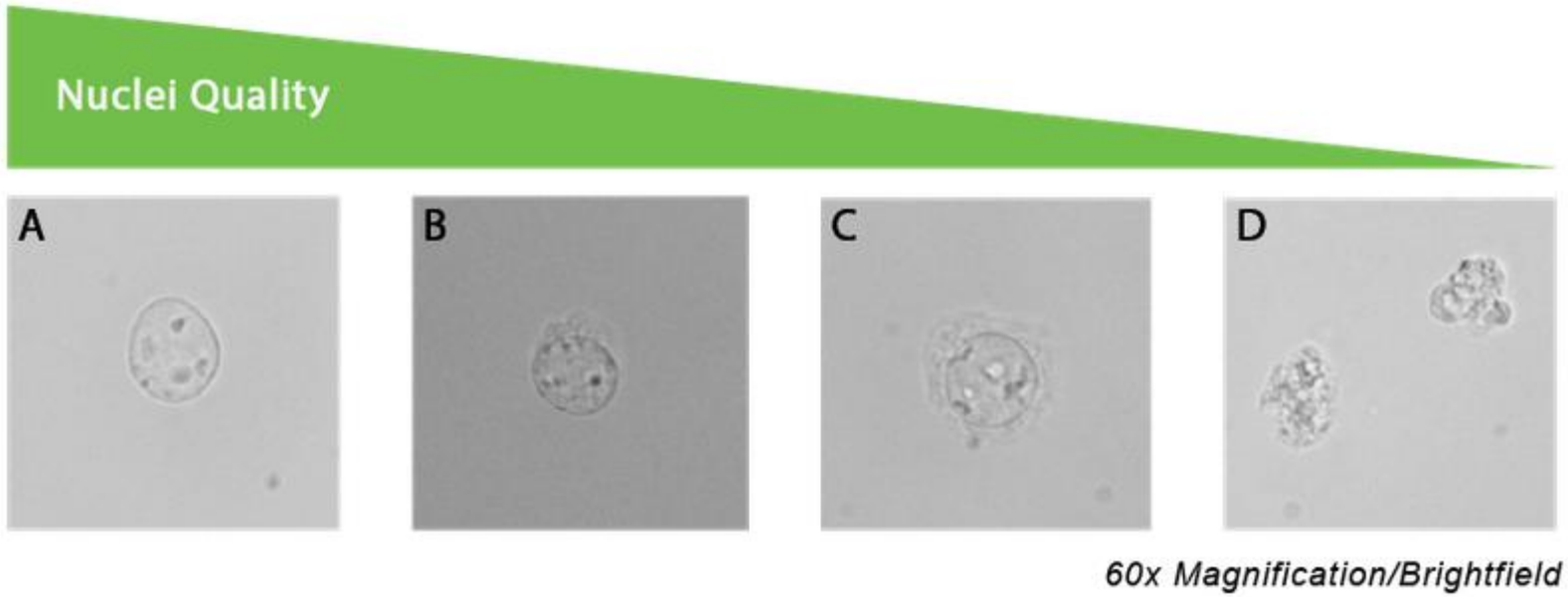


WTA Next enables sensitivity of >3,000 median genes/cell at 25,000 raw reads/cell from resting PBMCs samples

Single-cell sequencing – Know your cells

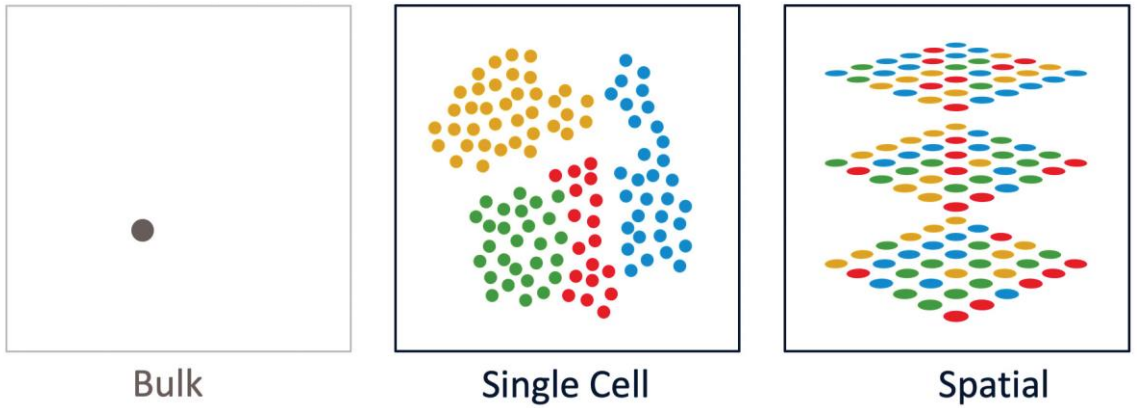
- Optimize your tissue dissociation to obtain high viability cell suspensions
- Check viability of sample over time, but don't waste your sample
- Work fast but not rushed
- Stagger your samples so they don't sit for too long
- Keep samples at 4°C when possible
- Keep cells in the buffer where they are the happiest up until cell capture
- If viability decreases over a short period of time this will be reflected in transcriptional data
- Poor viability → high % mitochondrial counts and high background due to free floating RNA
- Number of trypan positive cells x number of wasted sequencing reads
- Dying cells can pass QC and be regarded transcriptionally 'healthy'

Single-cell sequencing – example nuclei quality



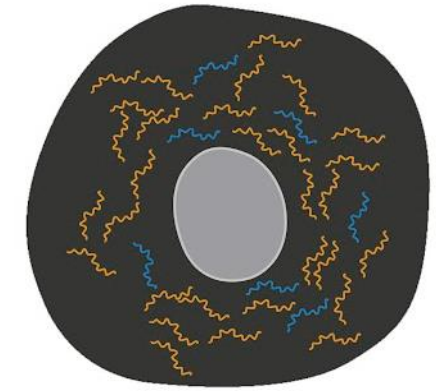
Single-cell – types

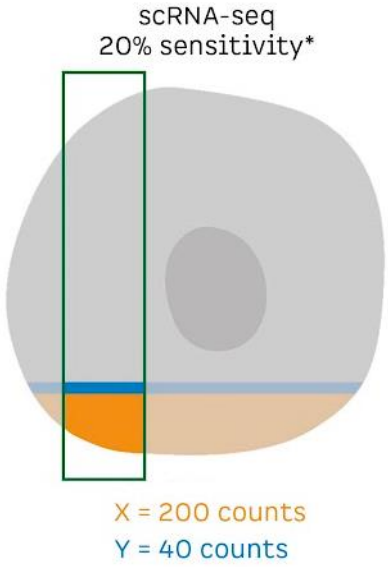
Different types of RNA-seq data

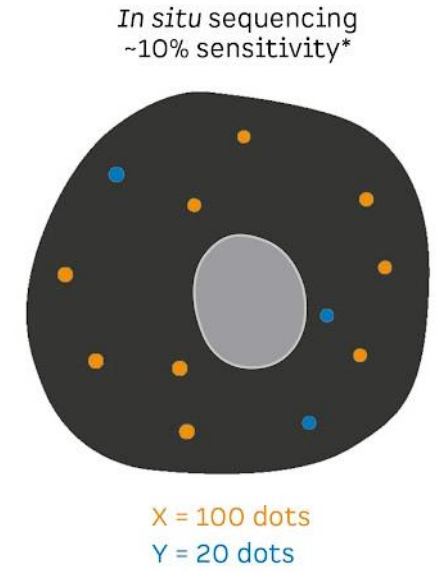


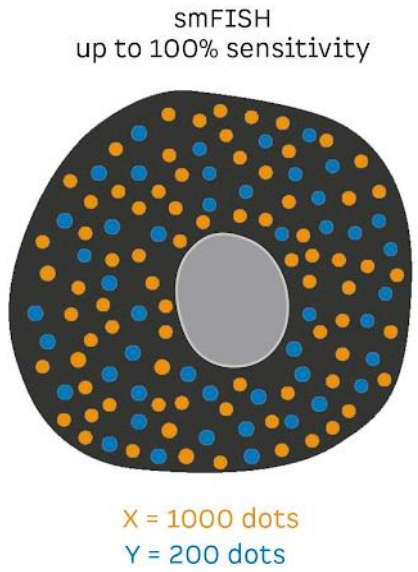
- **Target** -> In situ or transfer mRNA to slide
- **Source of data** -> Probes or full cDNA
- **Detection** -> Decoding probes or NGS

Different types of Spatial data







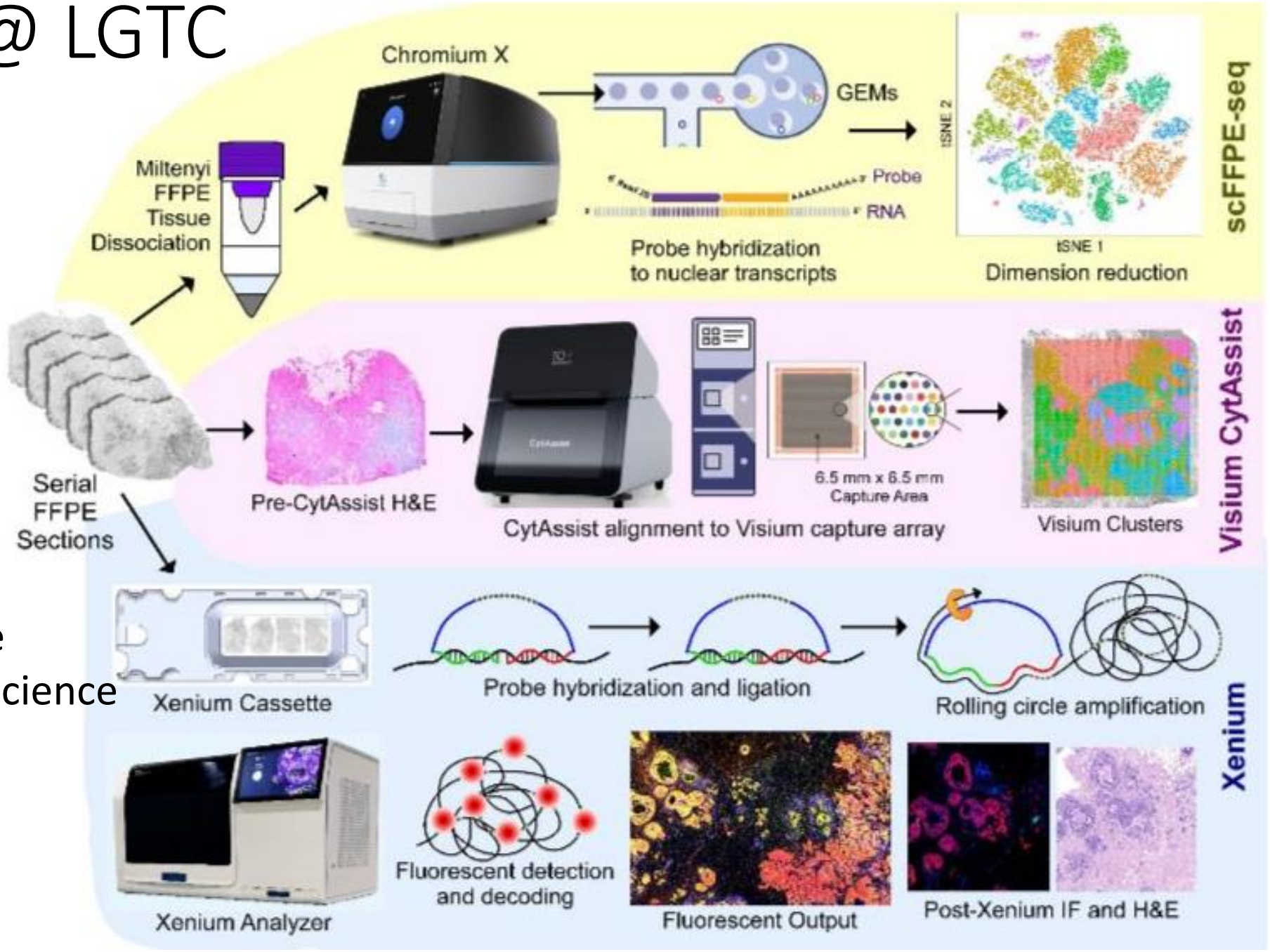


Single-cell @ LGTC

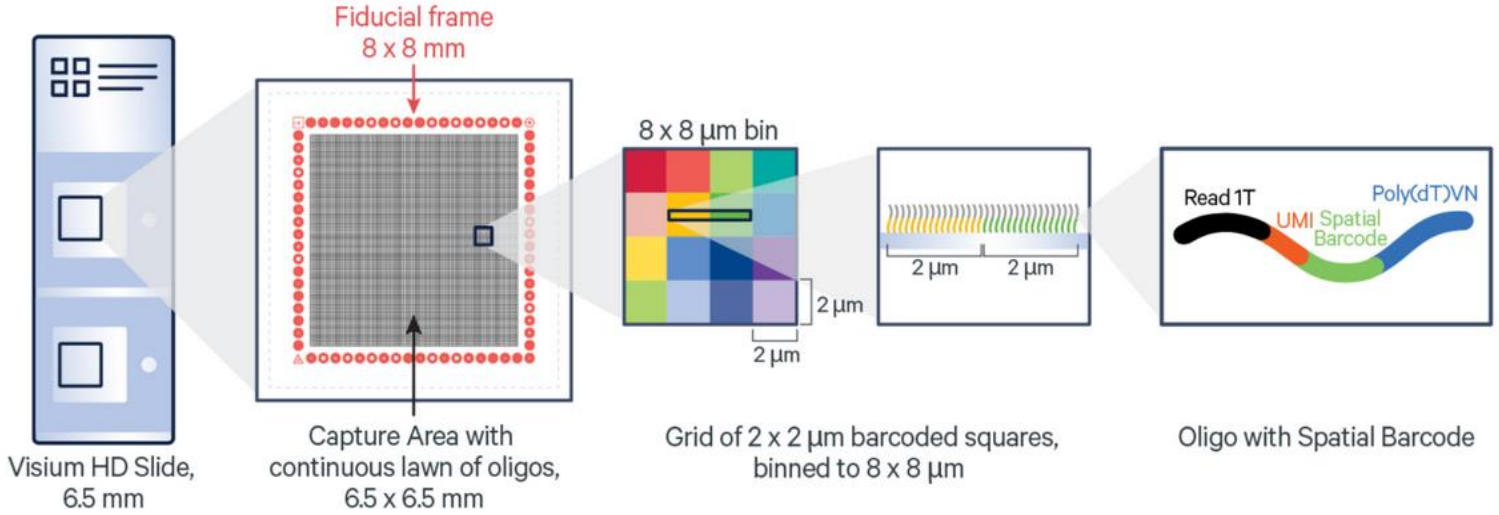
10xGenomics
- Chromium X
- CytAssist
- Xenium

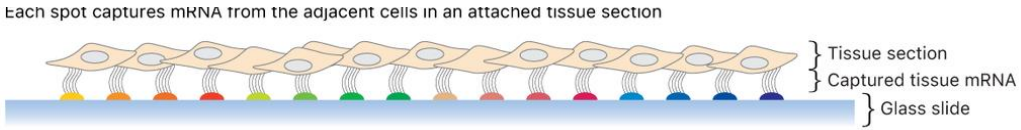
BD Rhapsody

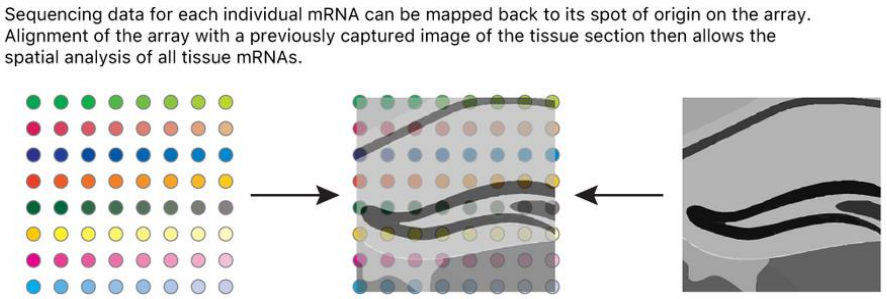
Exploring:
- Elemental Bioscience
- PipSeq -> Fluent Bioscience
- Atrandi Biosciences

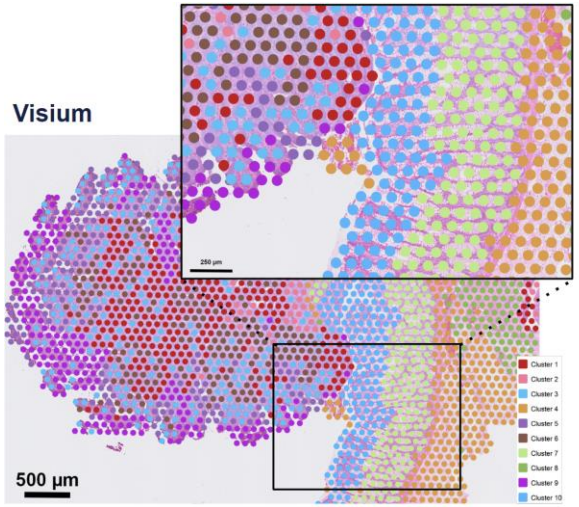


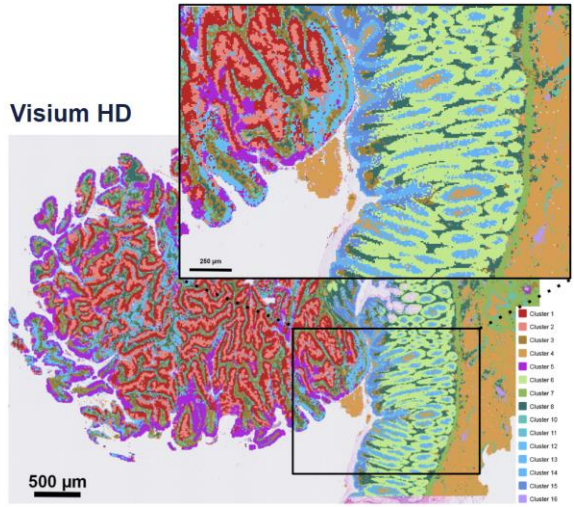
Single-cell – Visium CytAssist







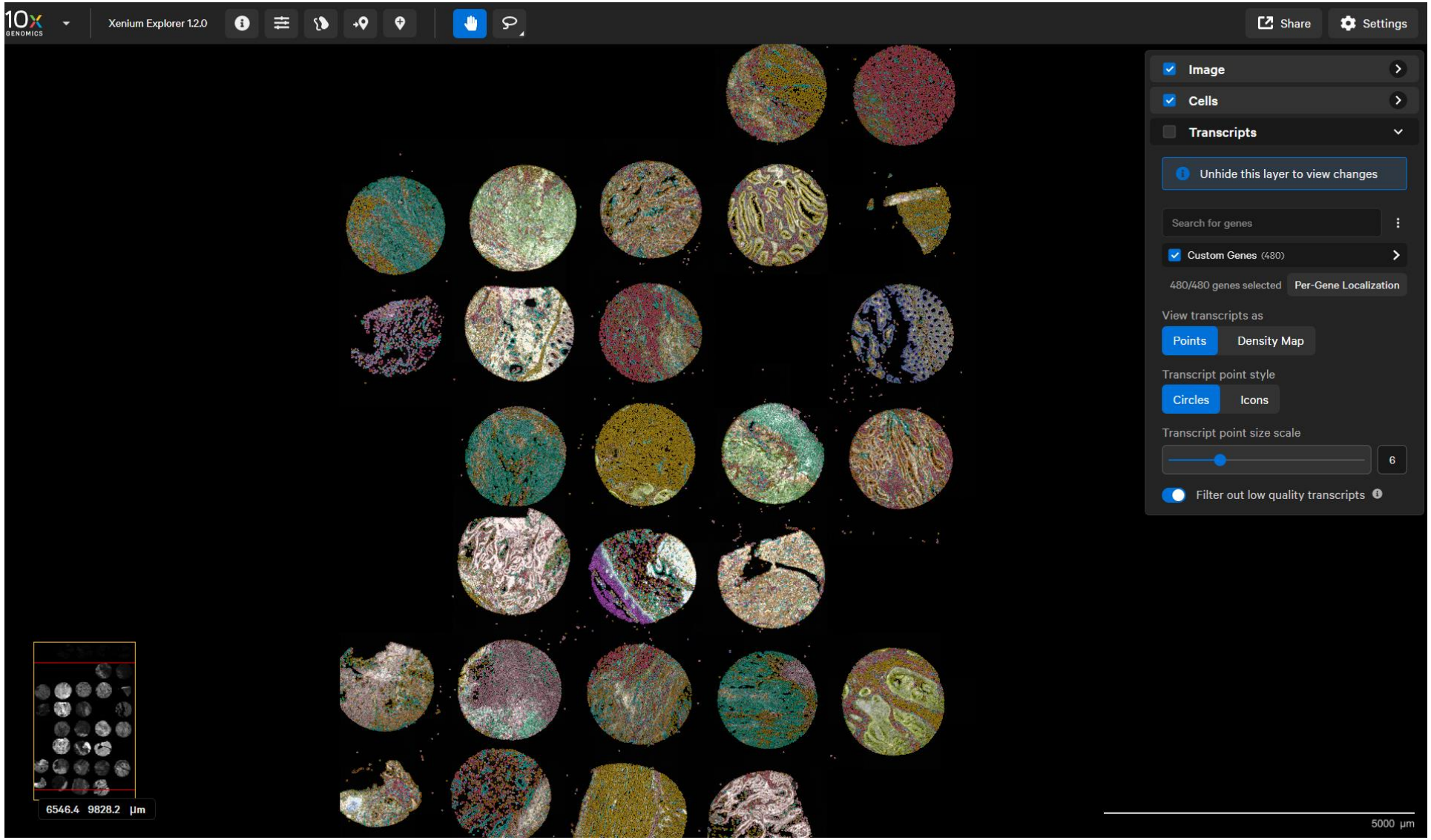




Transfer either the **mRNA** or **hybridized probes** to surface of slide

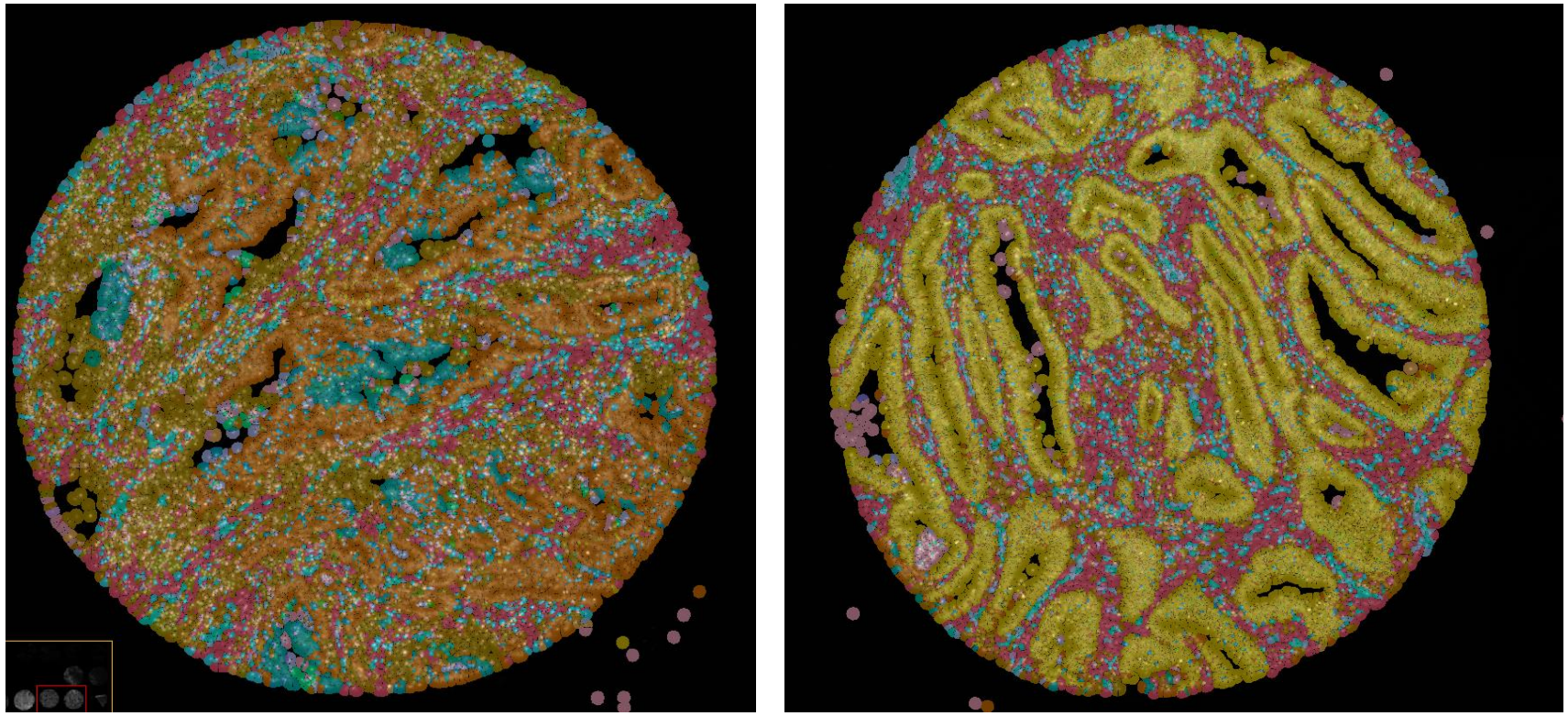
- 10x Genomics Application Note

Single-cell – Xenium – Tissue Array



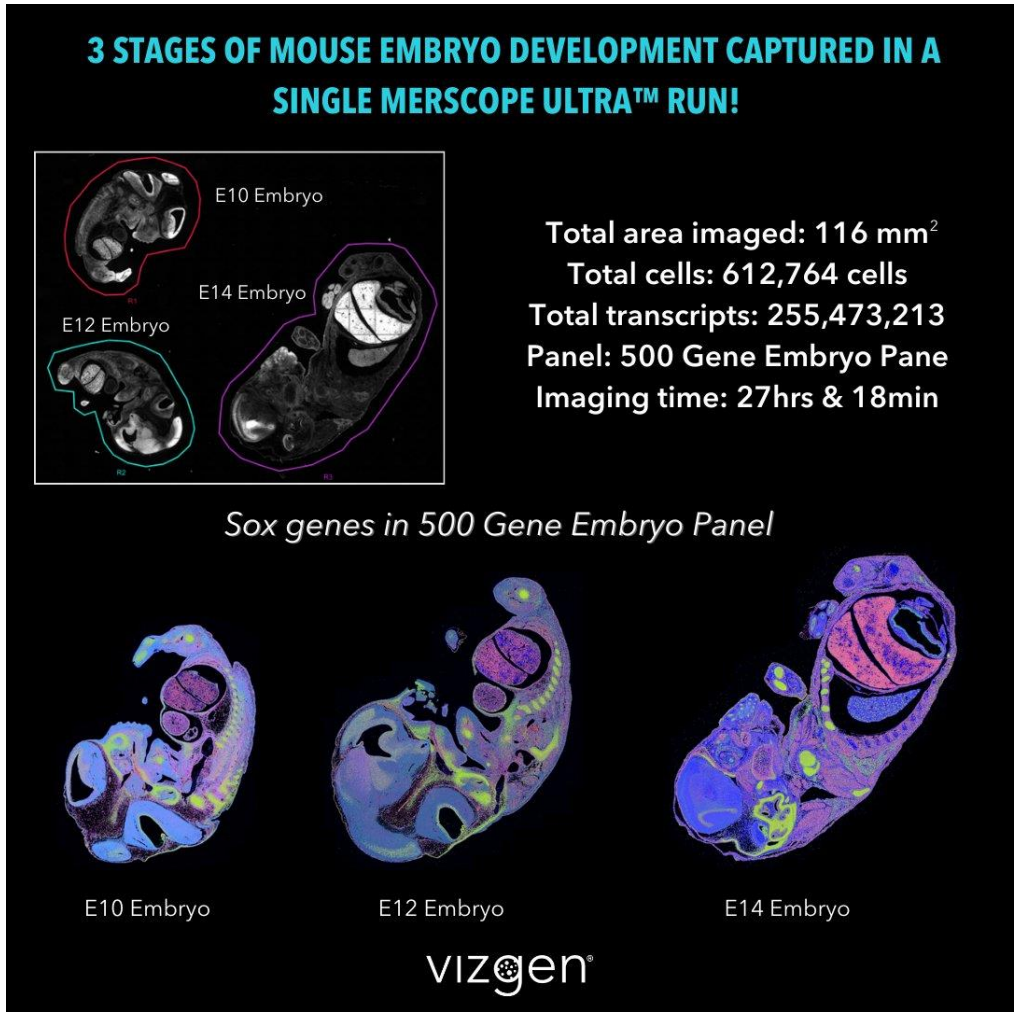
Detected number of genes: 100-5000

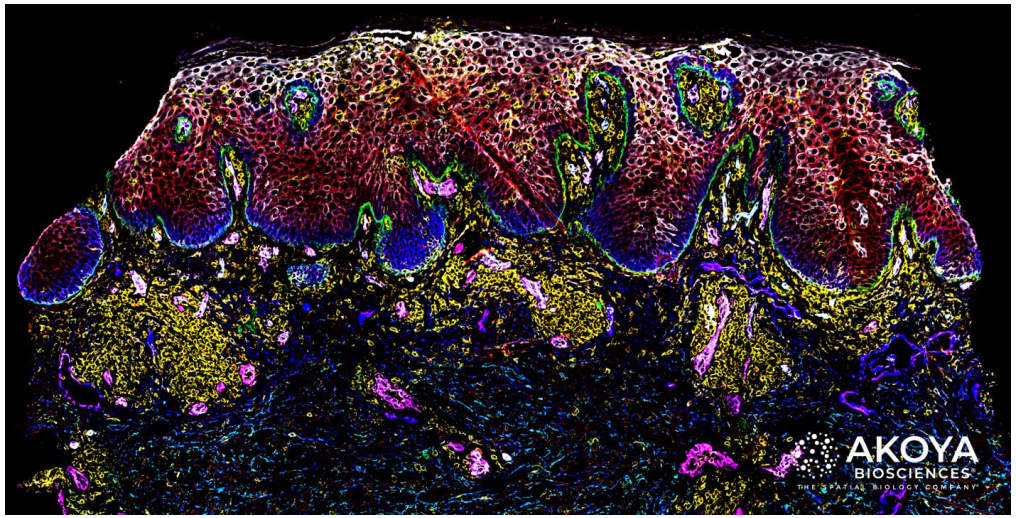
Single-cell – Xenium



- Connectivity of cells
- Sub-Cellular detection
- Challenging is cell segmentation

Single-cell – Competition for 10xGenomics

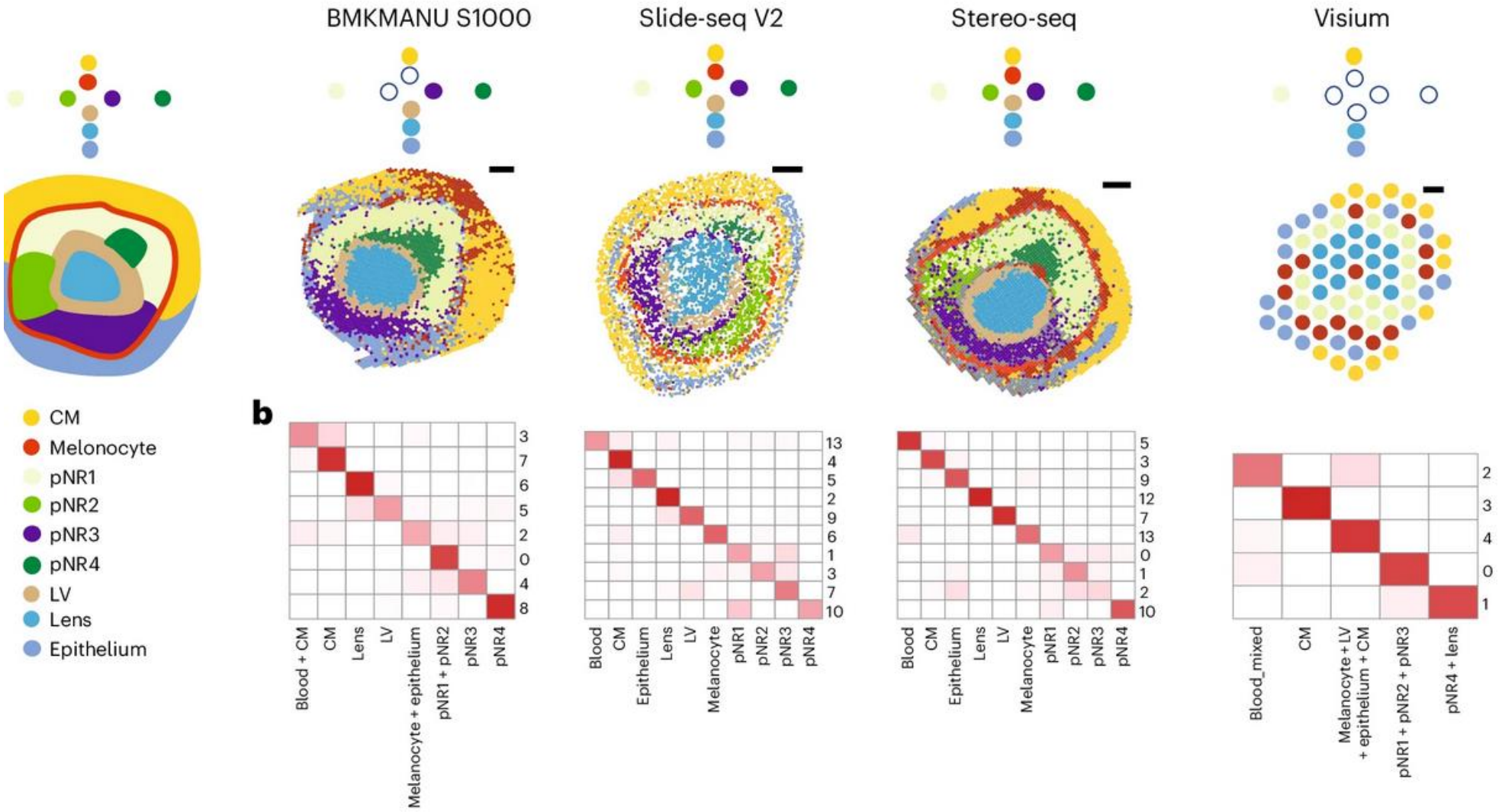






Alternatives for Spatial Transcriptomics

Capture **mRNA** and sequencing cDNA



Systematic comparison of sequencing-based spatial transcriptomic methods

You, Y., Fu, Y., Li, L. *et al.* Systematic comparison of sequencing-based spatial transcriptomic methods. *Nat Methods* **21**, 1743–1754 (2024). https://doi.org/10.1038/s41592-024-02325-3

Conclusions

- The vendor didn't use your sample to produce data
- Think before you start
- Junk in, junk out
- QC is VERY important.
- If you’re unsure, ask for support! (info@lgtc.nl)

Helpful sample prep links

- DNA library prep overview
- Assessment of nucleic acid quality
- Review of library prep challenges
- Illumina support page detailing sample-specific and extraction-specific contaminants